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EART11100

How to Build a Planet

Week 9

Tectonics and climate change across the Phanerozoic

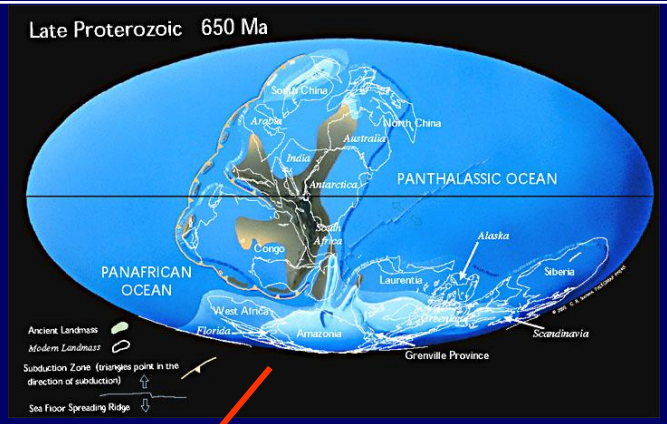
The Paleozoic Earth: the geological and climatic drivers behind the Cambrian Explosion to Permian Mass Extinction

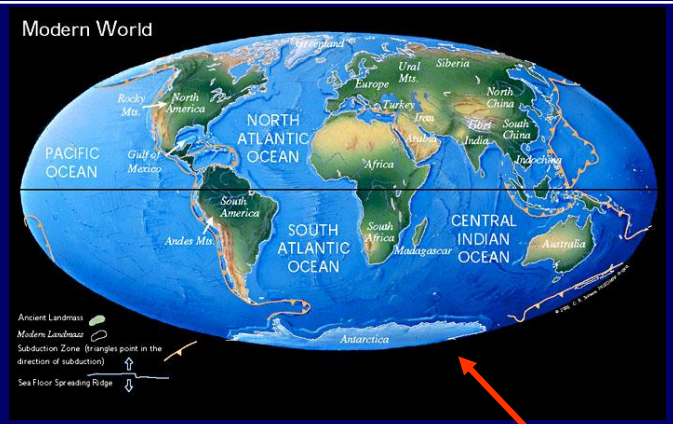
Learning objectives

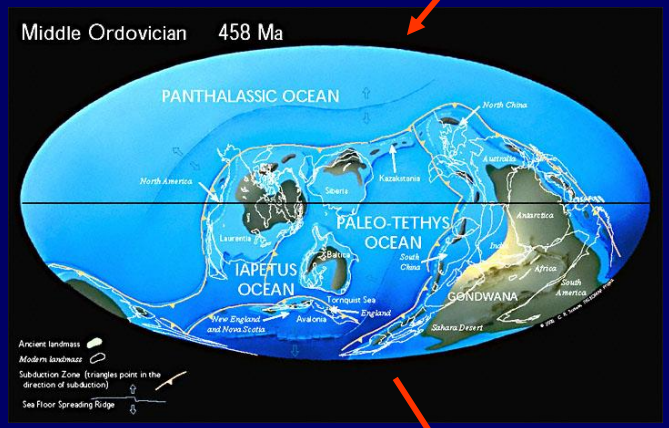
On successful completion of this week's class you should be able to:

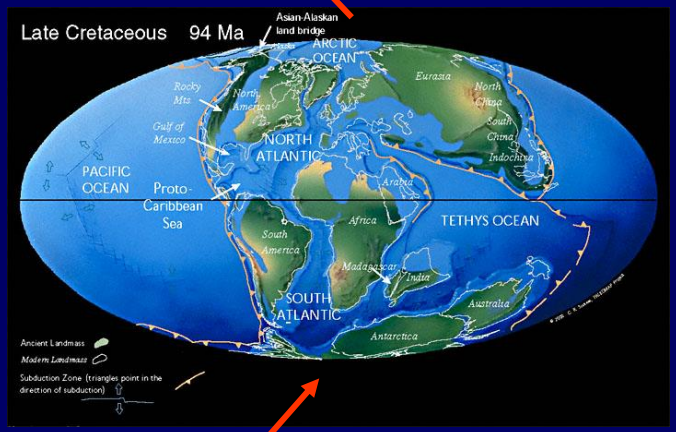
- name the major continents and oceans present during the Paleozoic and describe their dispersal and amalgamation through time
- draw a global sea level curve for the Phanerozoic and identify the periods when sea level was highest and lowest relative to today
- describe the palaeogeography and climate of the Earth during each period of the Paleozoic
- explain the interrelationship between the geosphere, hydrosphere and biosphere, using an example from each geologic period

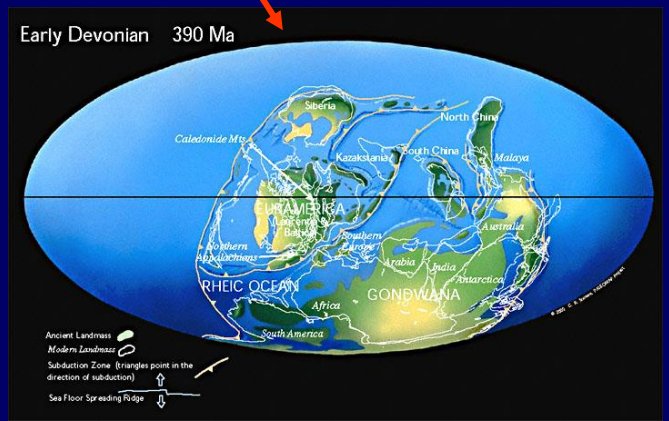
About the Phanerozoic palaeogeography: Pangaea

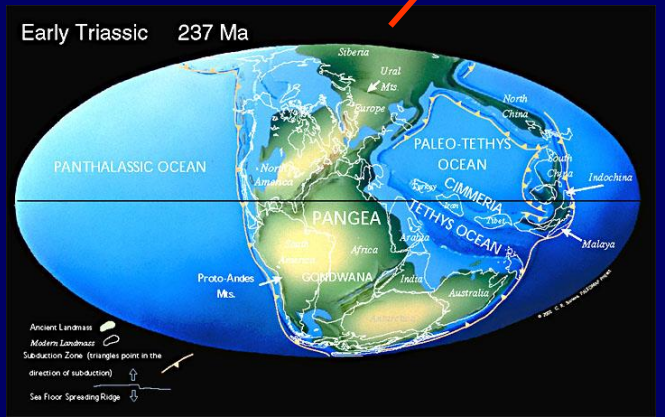






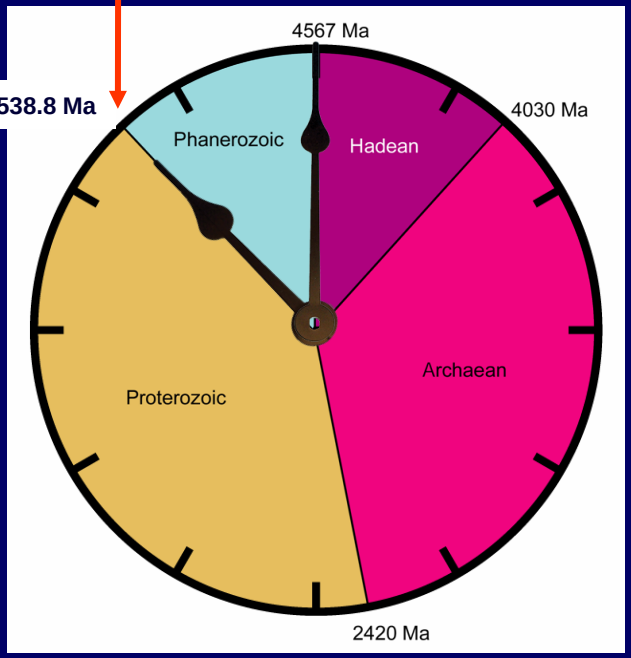






About the Phanerozoic: Greek ‘visible life’

Cambrian ‘explosion’ or ‘evolutionary Big Bang’



	Eon	Era	Period	Epoch	
Younger	Phanerozoic	Cenozoic	Quaternary	Holocene	present
				Pleistocene	
			Neogene	Pliocene	
				Miocene	
			Paleogene	Oligocene	
				Eocene	
				Paleocene	66.0 Ma
		Mesozoic	Cretaceous	~	
			Jurassic	~	
			Triassic	~	251.9 Ma
		Paleozoic	Permian	~	
			Carboni-ferous: Pennsylvanian	~	
			Carboni-ferous: Mississippian	~	
			Devonian	~	
			Silurian	~	
			Ordovician	~	
			Cambrian	~	538.8 Ma
Older	Proterozoic	~	~	~	
	Archean	~	~	~	
	Hadean	~	~	~	

Phanerozoic Eras

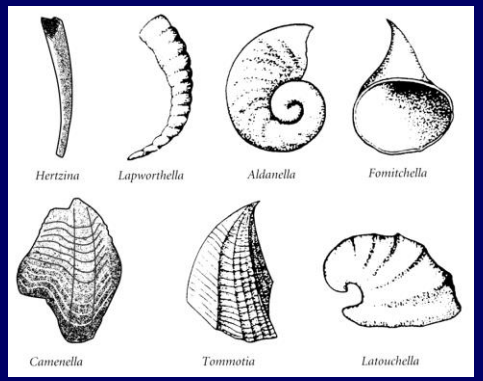
Paleozoic – ancient life

Mesozoic – middle life

Cenozoic – recent life



Treptichnus pedum – earliest complex trace fossil first appearance defines base of Cambrian

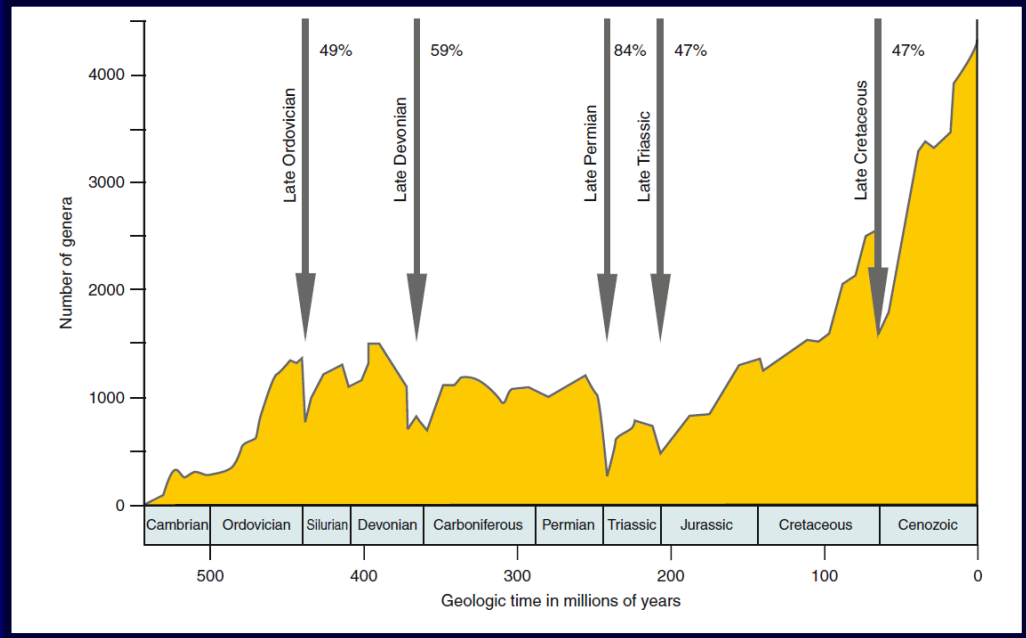


Small shelly fossils (<a few mm)

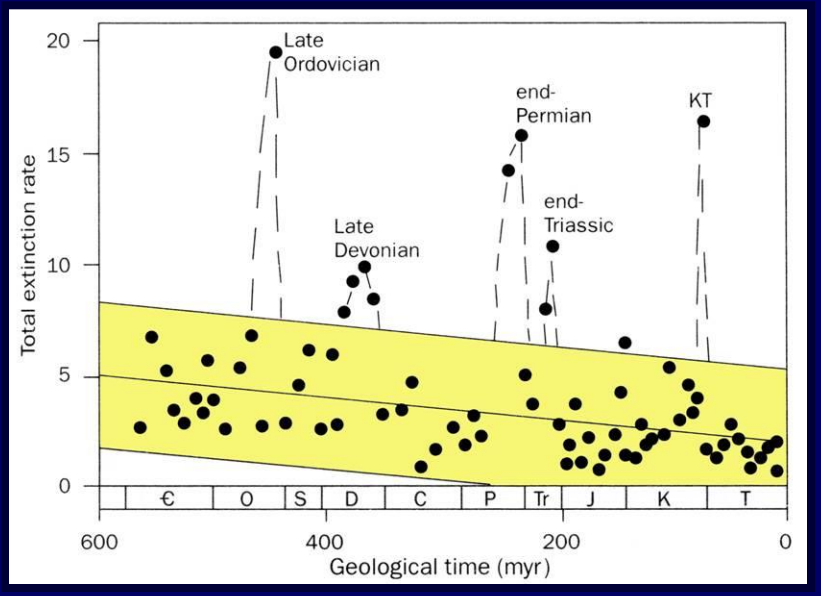
Key biotic events (mass extinctions) of the Phanerozoic

Mass Extinction event definition

- 30%+ species become extinct
- affect a wide variety of ecologies
- worldwide
- within a short period of time



Assumed % of genera extinct during major extinction events

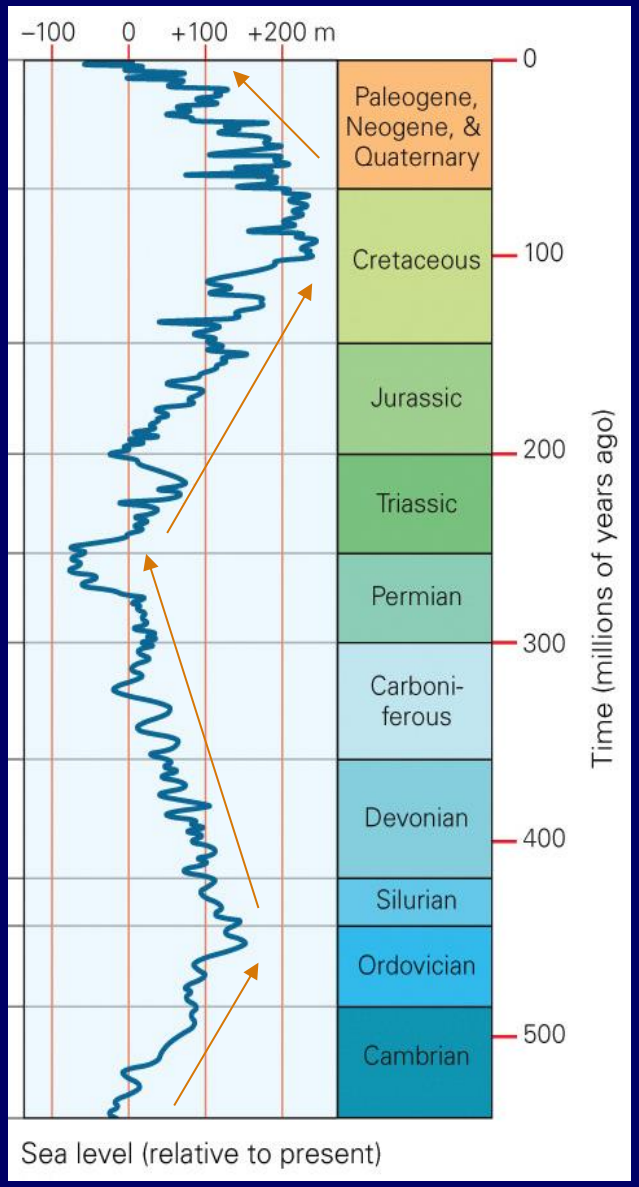


Marine invertebrate extinction record (extinctions/Ma)

Causes:

- meteorite impacts
- major global climate changes
- periods of major igneous activity

Phanerozoic global sea level changes



Opening of Atlantic

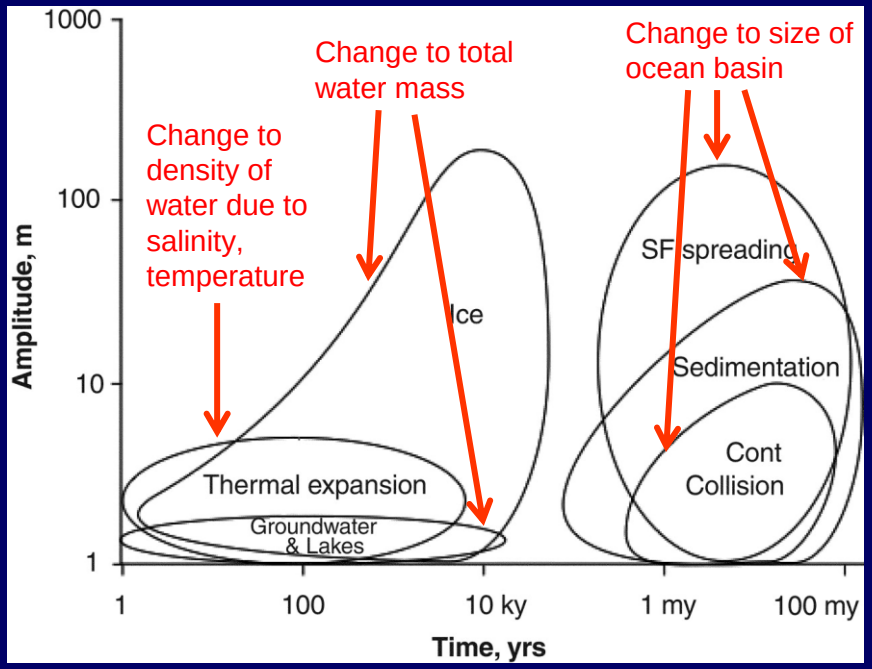
Pangea

Supercontinent assembly leads to:

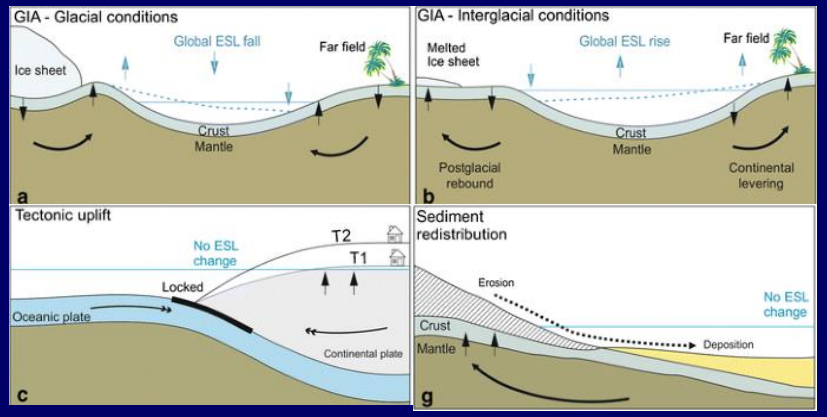
- thickening of continents during orogeny decreasing continental area resulting in increased oceanic area
- and/or higher elevations that result when trapped heat builds up below the supercontinents

Break up of Rodinia

Causes of eustatic & regional sea level change



Timing and amplitudes of geologic mechanisms of sea level change



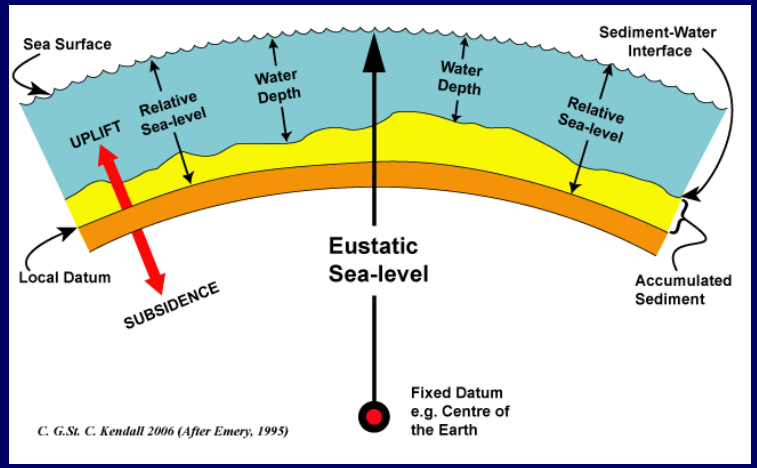
An increase in relative sea level (the balance between changes in eustacy, subsidence rate + rate of sediment supply) → flooding of continents

Eustatic sea level (ESL) changes in mass and volume of oceans

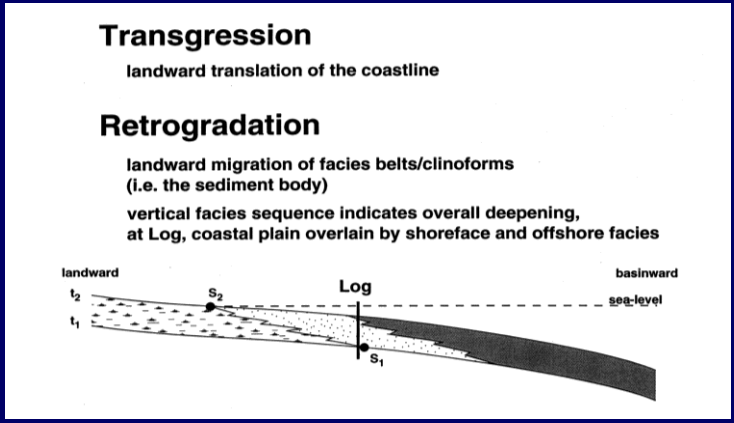
(1) climatic changes in continental water storage (as liquid or ice), ocean salinity or temperature

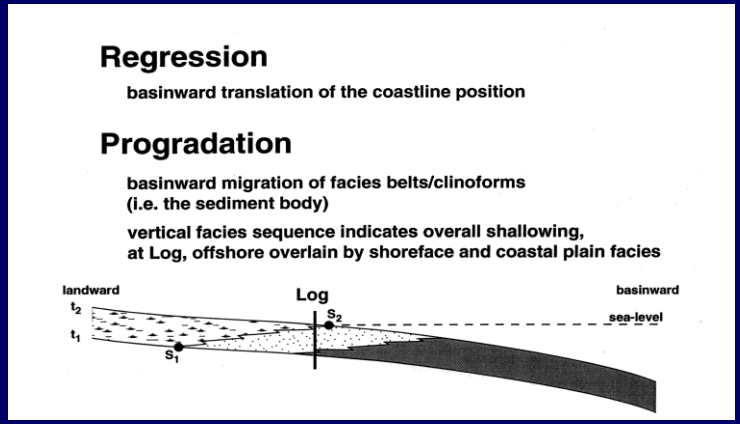
(2) geological forces changing ocean basin volume (continental collision, sedimentation or sea floor spreading)

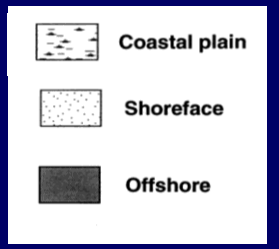
Relative sea level change = land uplift and subsidence (vertical tectonics) does not affect ocean water volume and causes relative sea level changes that vary over a range of spatial and temporal scales



Indicators of Phanerozoic sea level change







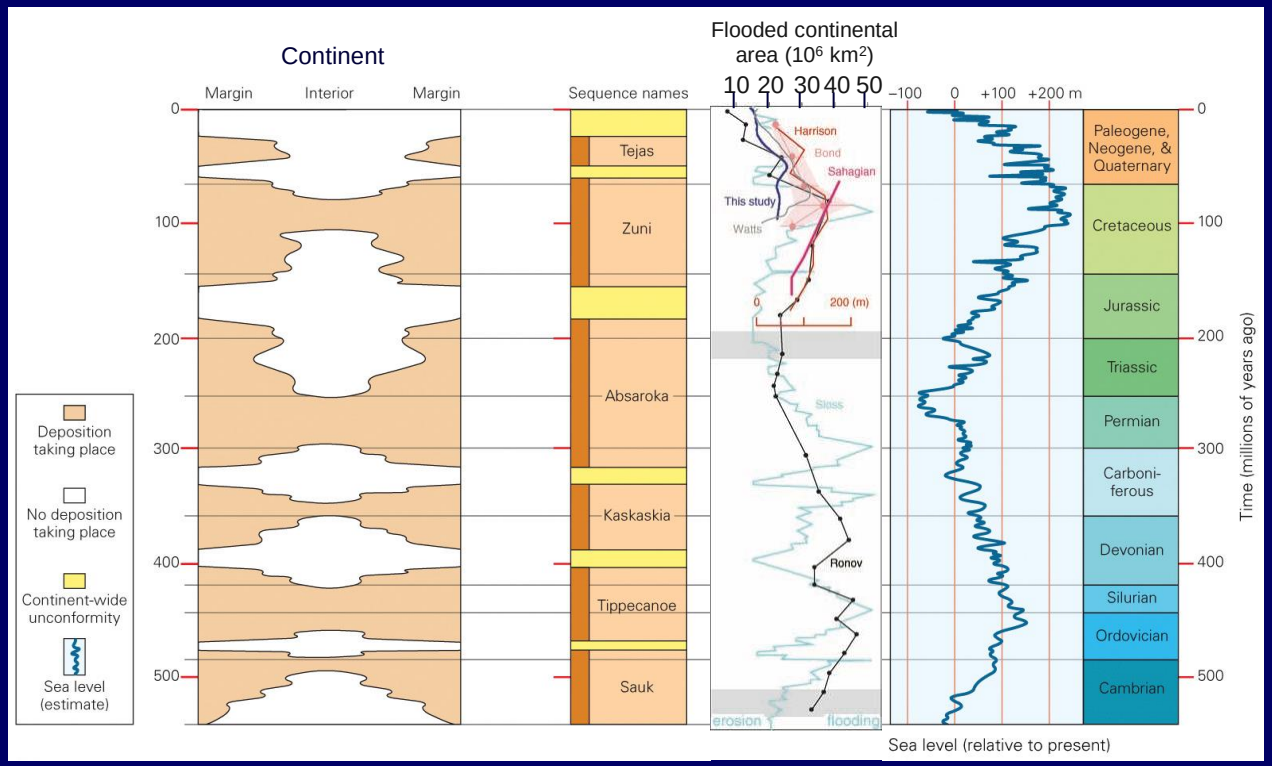
- When sea levels rise the shoreline migrates inland and the low-lying coastal plains or margins of the continents are submerged. These times of rising sea level are called **transgressions** and during these periods shallow marine sediments buried the continental deposits.

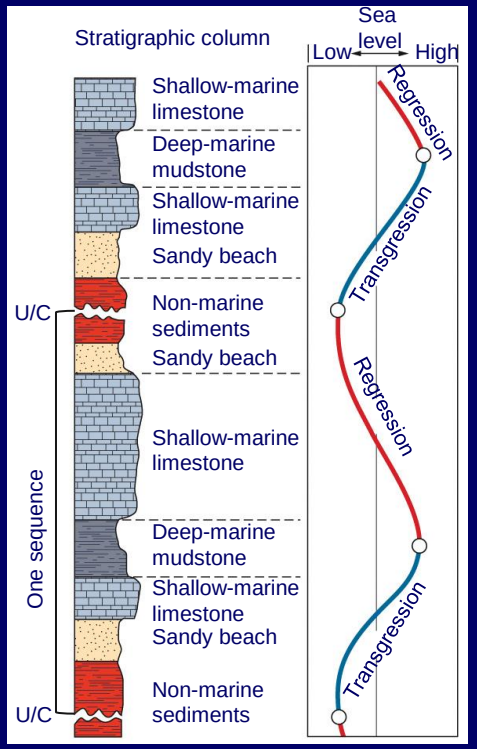
- Falling sea levels are called **regressions** and regional unconformities develop due to erosion or non-deposition.

Evidence for global eustatic variations:

- satellite measurements (<10 yrs),
- tide gauges (<150 yrs)
- shoreline markers - tropical reefs and atolls formed by shallow dwelling corals (< few 100 ka)
- oxygen isotopes ($\delta^{18}O$) of foraminifera (<200 Ma)
- Similar ages of unconformities on different continents have been interpreted as indicating that the surfaces were caused by a global process, eustasy (long term Phanerozoic record)

Continental flooding events through the Phanerozoic

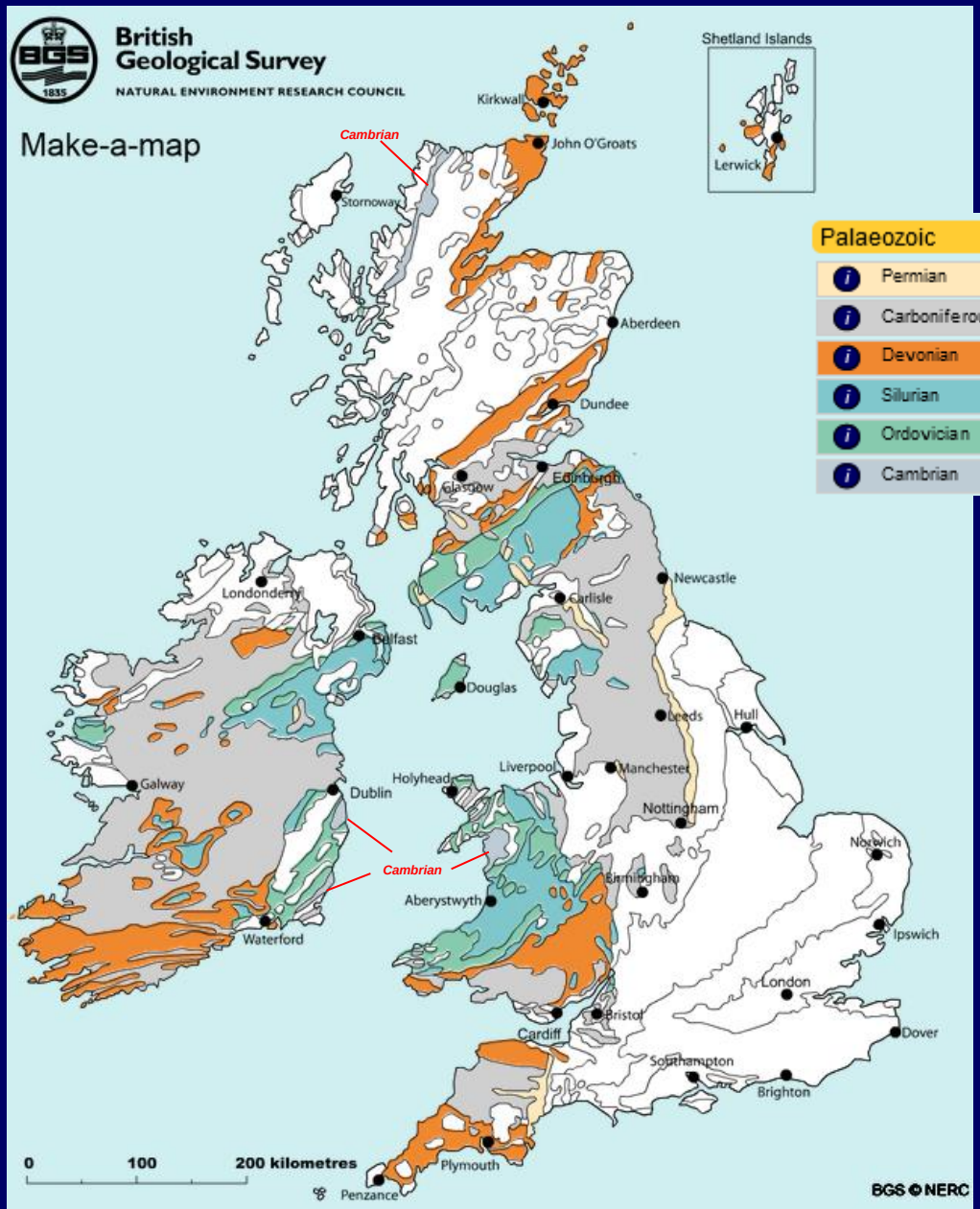




U/C = unconformity

- 10-100 Ma scale “supersequences” are attributed to global sea-level changes
- Sea-level changes on very long time scales (250 Ma) are related to the assembly and breakup of supercontinents

Palaeozoic rocks in the UK



Role of base Cambrian unconformity and transgression

Base of Cambrian is marked by an unconformity with intense weathering

- Rodinia supercontinent sitting high in Neoproterozoic
- extreme weathering in hot periods between Neoproterozoic snowball Earth events (high CO_2)

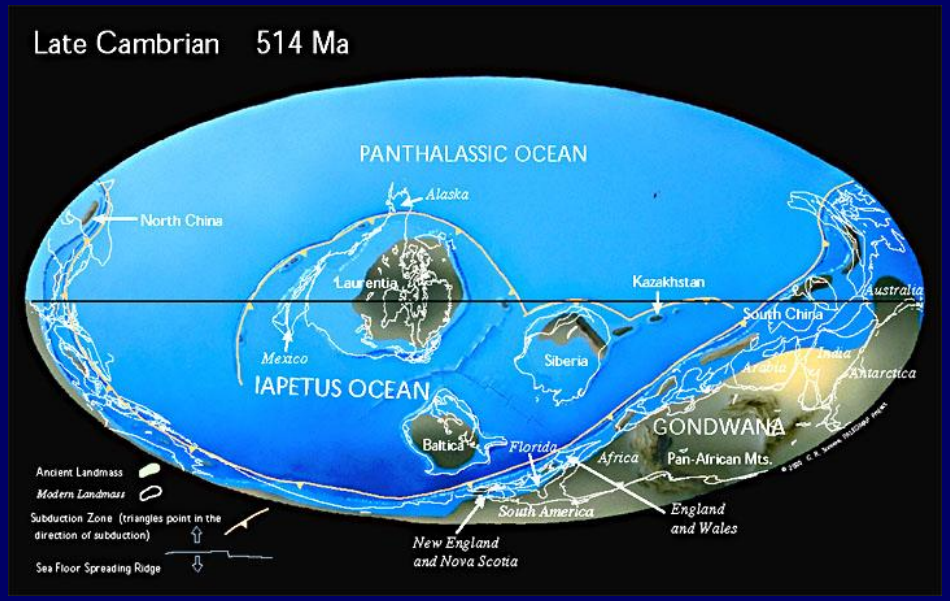
Break up of Rodinia → rising sea level → transgression → erosion → transportation of metals (e.g., Ca^{2+}) to oceans → oceans anonymously metalliferous

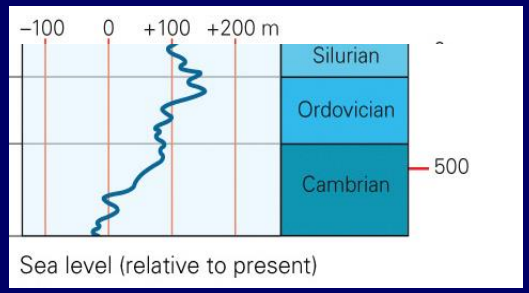
Solute-rich early Cambrian oceans (Ca^{2+} rich) → emergence of mineralized hard parts (shells, skeletons, *etc.*) in contemporary organisms

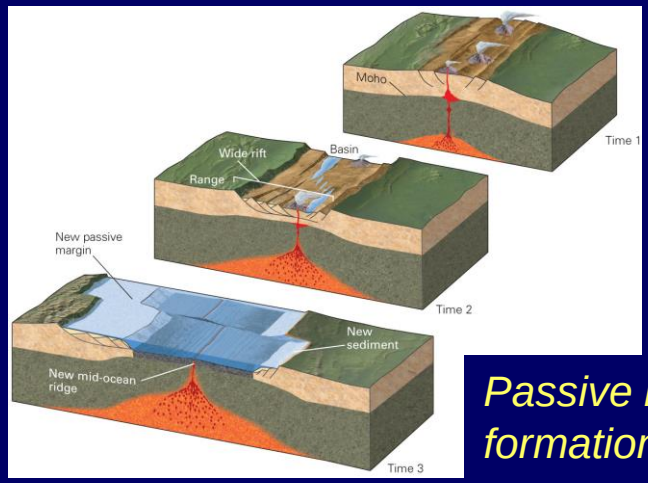


The Cambrian-Lewisian unconformity, Arkle, Sutherland

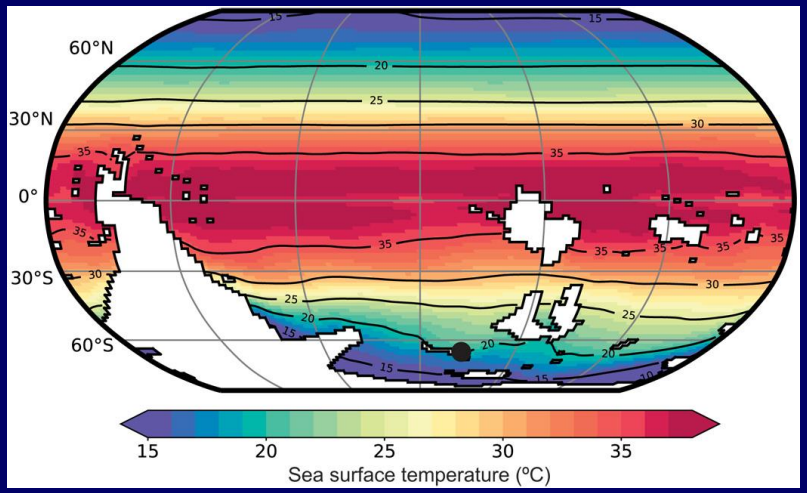
Break up of Rodina and rising sea levels







Passive margin formation



Cambrian sea surface temperatures

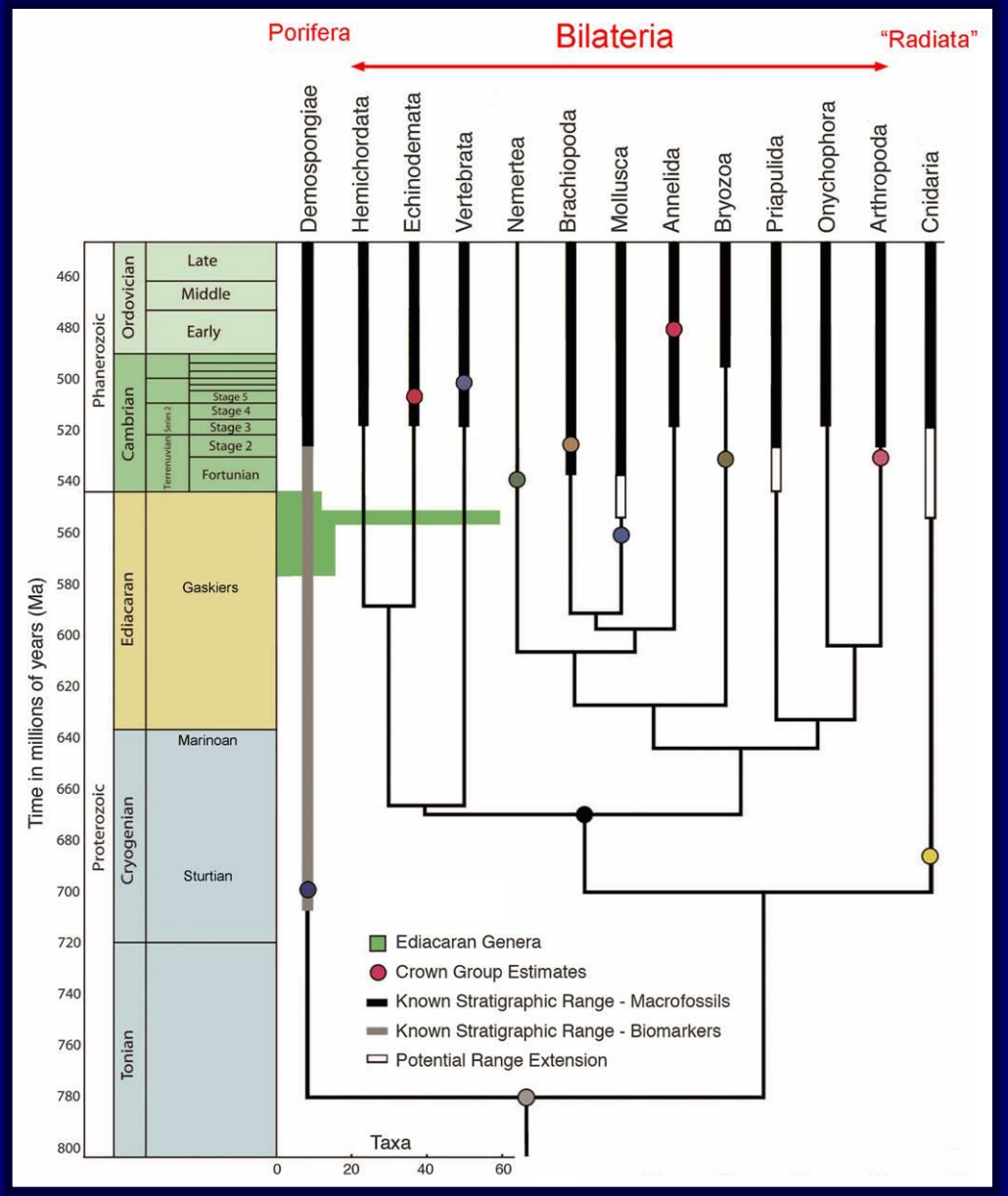
- Early Cambrian geological data support a greenhouse climate state lacking permanent polar ice sheets:
  - evaporite and calcrete deposits spanning a wide paleolatitudinal range,
  - deposition of tropical soils (laterites) at high paleolatitudes,
  - maximum of continental weathering rates over the past 900 Ma

The Cambrian Explosion: causes

What factors led to the building of large, complex bodies, the construction of mineralized skeletal material, and the development of metabolically expensive strategies (*e.g.*, burrowing, predation)?

→ environment - ocean salinity changes (esp. Ca^{2+} ions) due to deep weathering, increased atmospheric and ocean oxygenation (NOE)

→ ecology - feedbacks from successful evolutionary innovations incl. predation, mobility



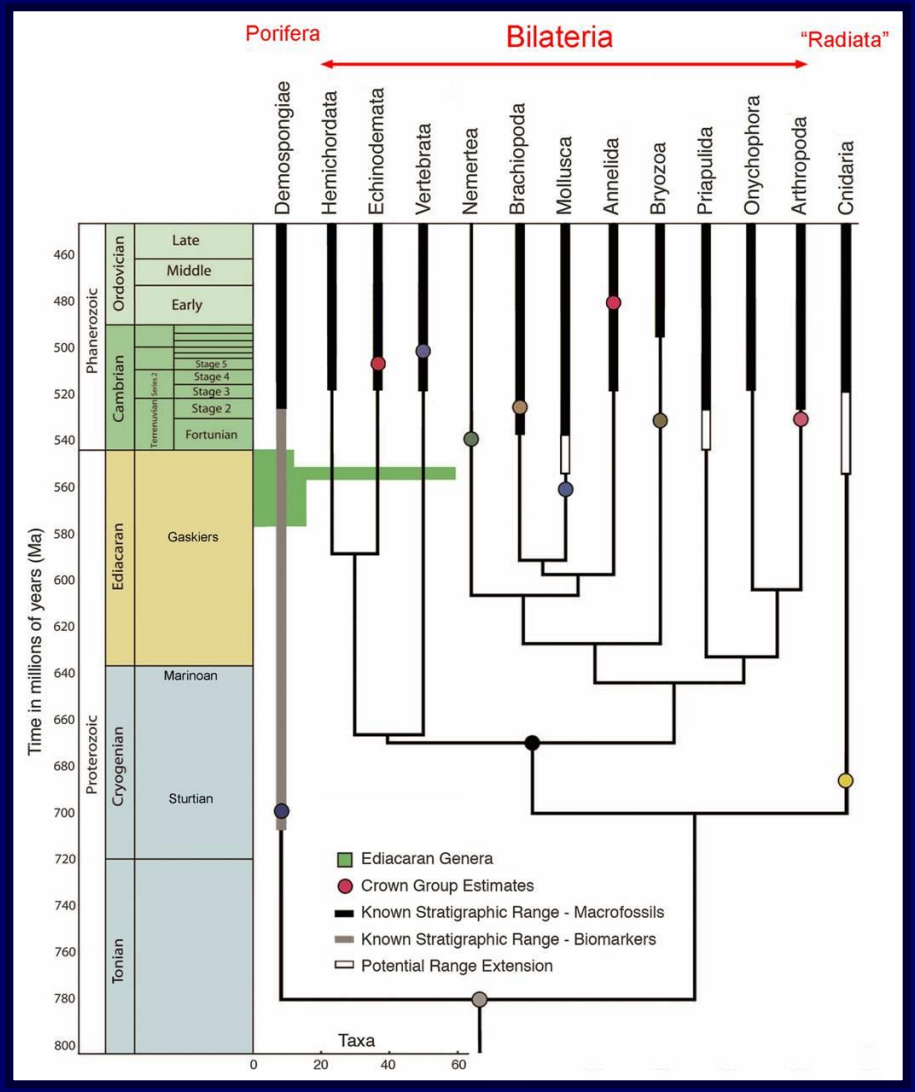
The advent of mineralised body parts (biomineralization)

‘Worm- World’: In the late Ediacaran minute conical tubular shells from worm-like organisms appear in the fossil record (advent of biomineralization)

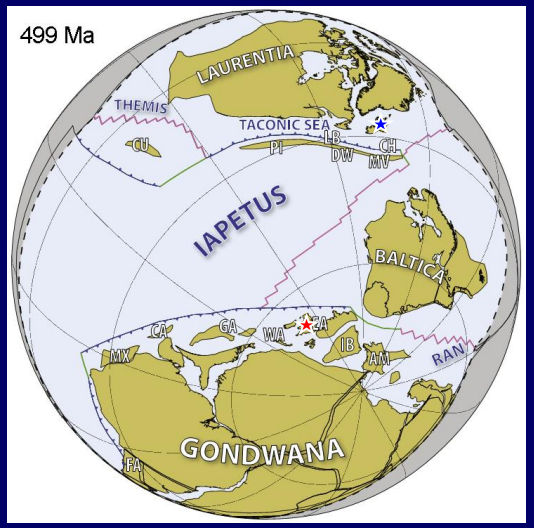
Biomineralization occurred *independently* among diverse bilaterians and other metazoans in the Cambrian
i.e. not just one ancestral species

Small biomineralized parts (either whole tiny skeletons or individual sclerites [armour pieces] on larger animals) were easier to construct first, and complete larger skeletons only happened later.

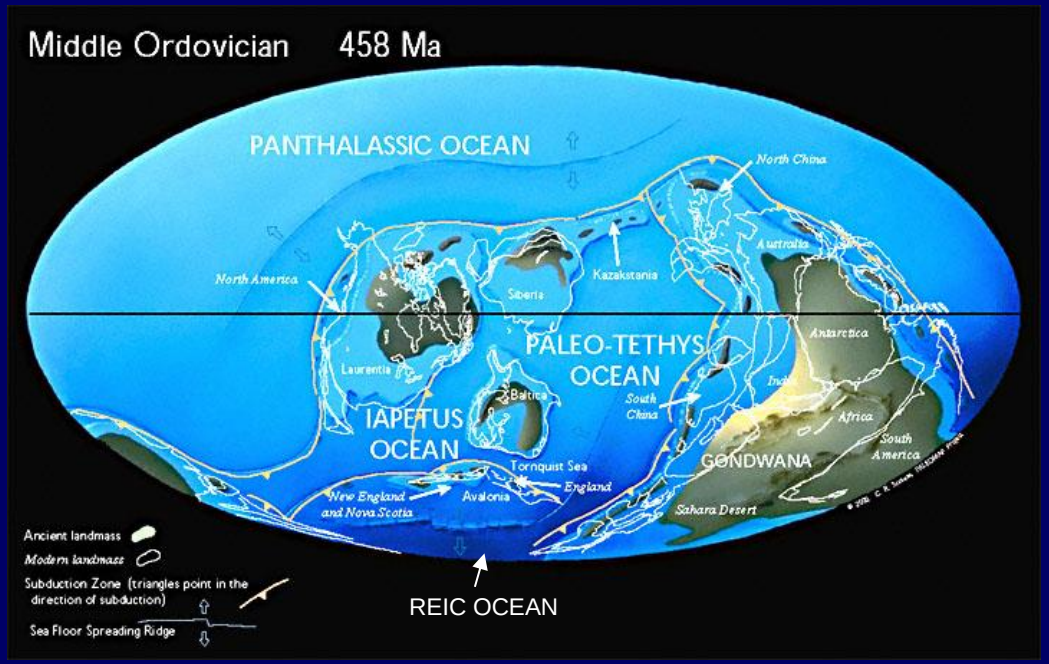
Thick black lines are the fossil records of each of the important animal lineages, most make their first appearance in the Cambrian.



Ordovician (485.4 – 443.8 Ma) palaeogeography

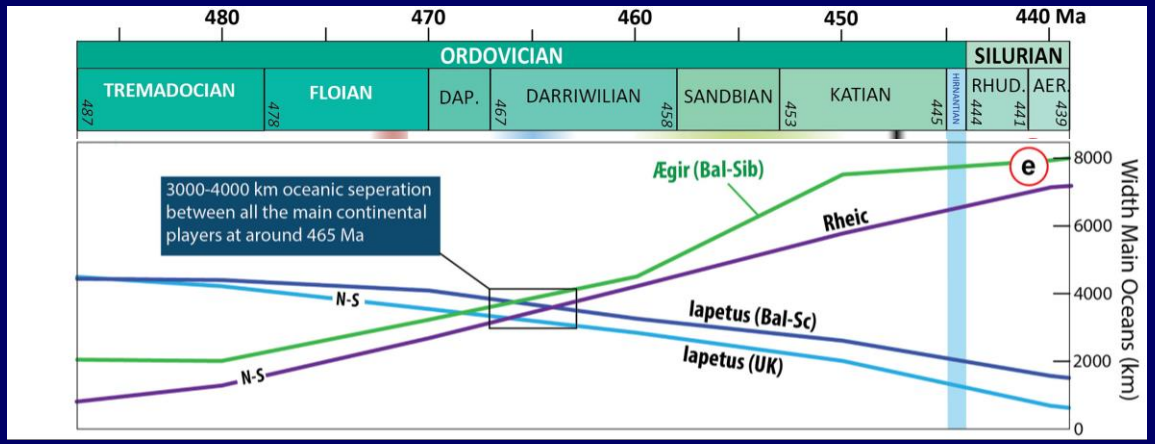


At the start of the Ordovician the Iapetus Ocean >4000km wide





Closing of the Iapetus Ocean and opening of the Reic Ocean

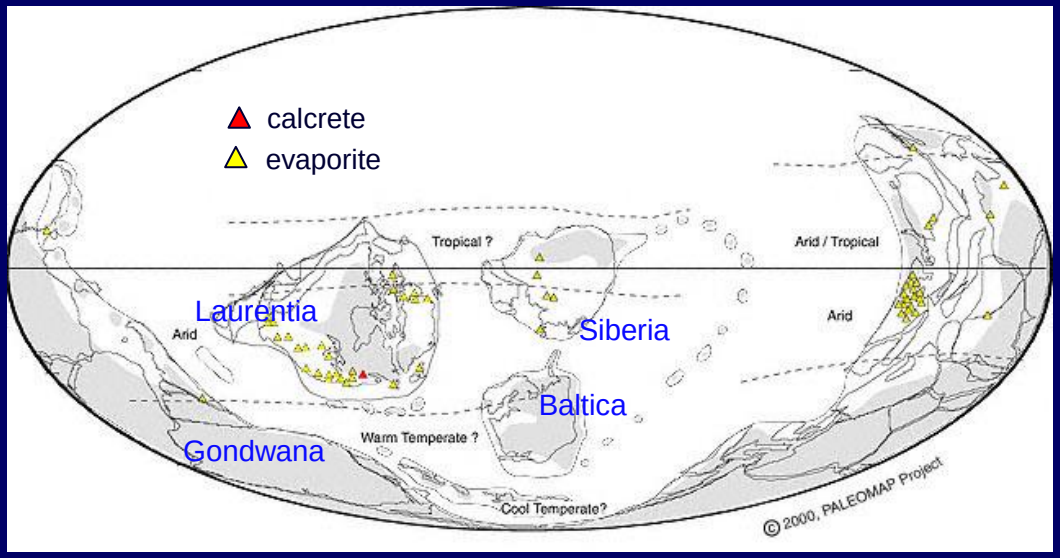


Changes in ocean widths during the Ordovician

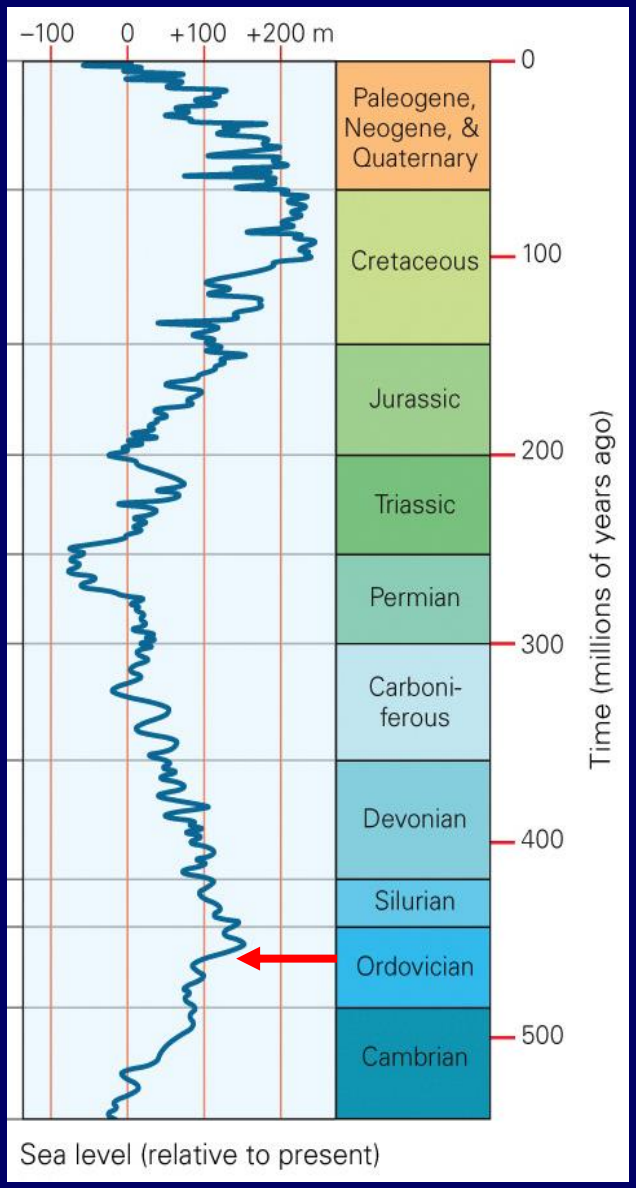
Ordovician sea level and climate

- Continent dispersal led to a pronounced transgression in mid-Ordovician creating large areas of tropical continental shelves on the margins of the continents.
- Throughout the Ordovician the climate was generally warm until a severe Late Ordovician glacial event.

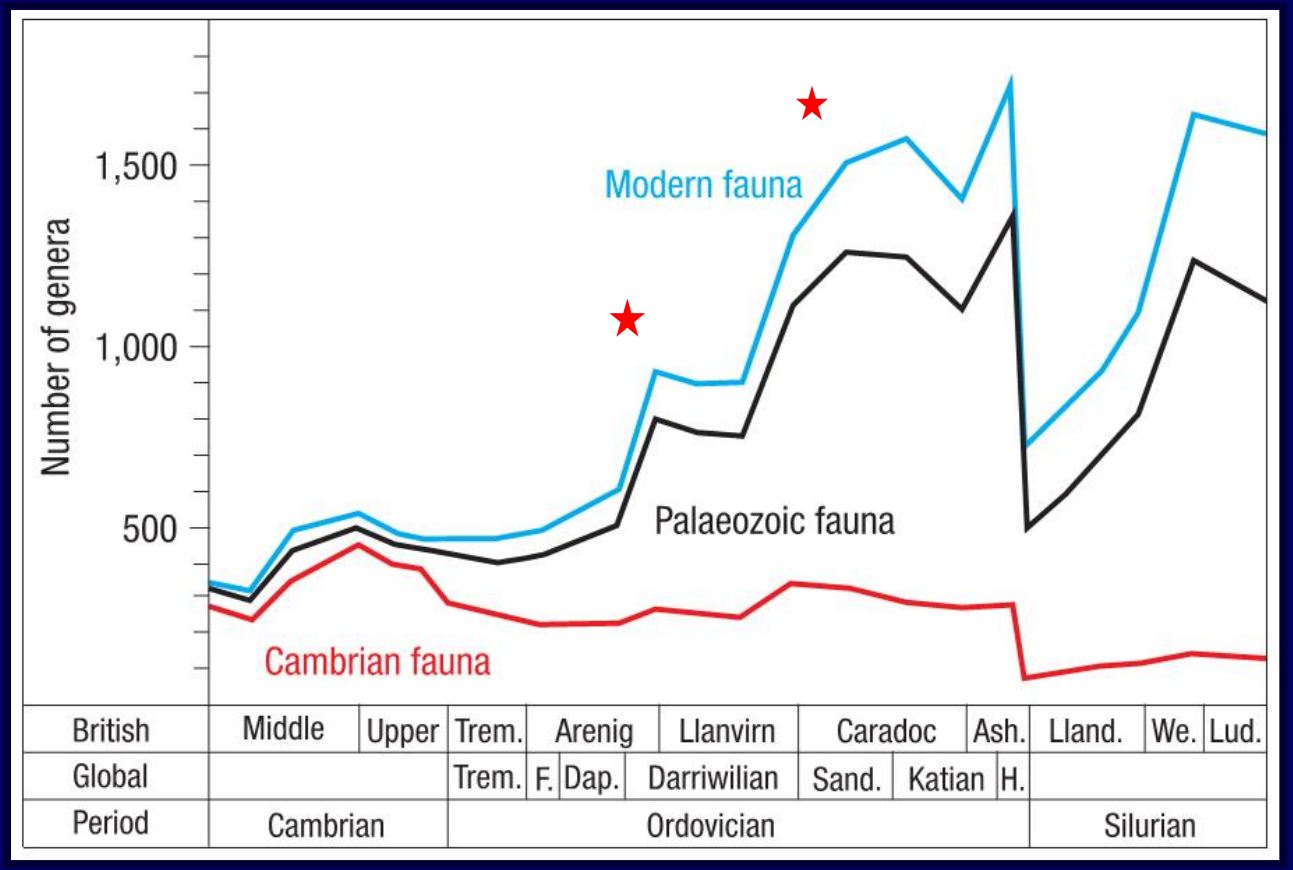
- **Calcrete** is the weathered crust in arid and semiarid regions, sand and silt cemented by calcite, dolomite, gypsum, halite, and ferric oxide.
- **Evaporite** largely bedded halite, sylvite, gypsum, anhydrite, and various potash salts form from the precipitation of solid mineral crystals from a concentrated solution of salt- or freshwater. Evaporite deposits may be either marine or nonmarine (lacustrine) in origin.



Lower Ordovician climatic zones

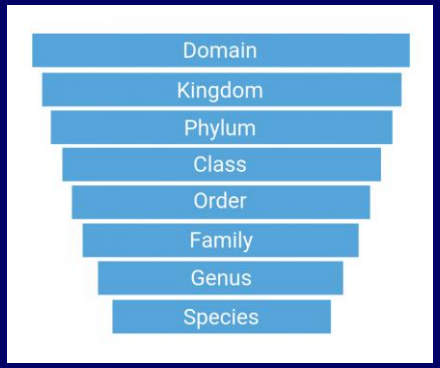


Great Ordovician Biodiversification Event (GOBE)



Cambrian Explosion many new phyla appeared however biodiversity at the family, genus and species level, remained low until the GOBE

The GOBE occurred in two bursts shown by stars



GOBE: Reefs and mud mound complexes

Period	Relative abundance	Dominant skeletal elements of reefs
Neogene		Corals, algae
Palaeogene		
Cretaceous		Rudist bivalves, corals, stromatoporoids
Jurassic		Corals, sponges, stromatoporoids
Triassic		Corals, stromatoporoids
Permian		Algae, sponges, corals
Carboniferous		
Devonian		Corals, stromatoporoids
Silurian		
Ordovician		Corals, bryozoa, stromatoporoids
Cambrian		
Precambrian		Stromatolites

Change from microbial- to metazoan-dominated reefs.



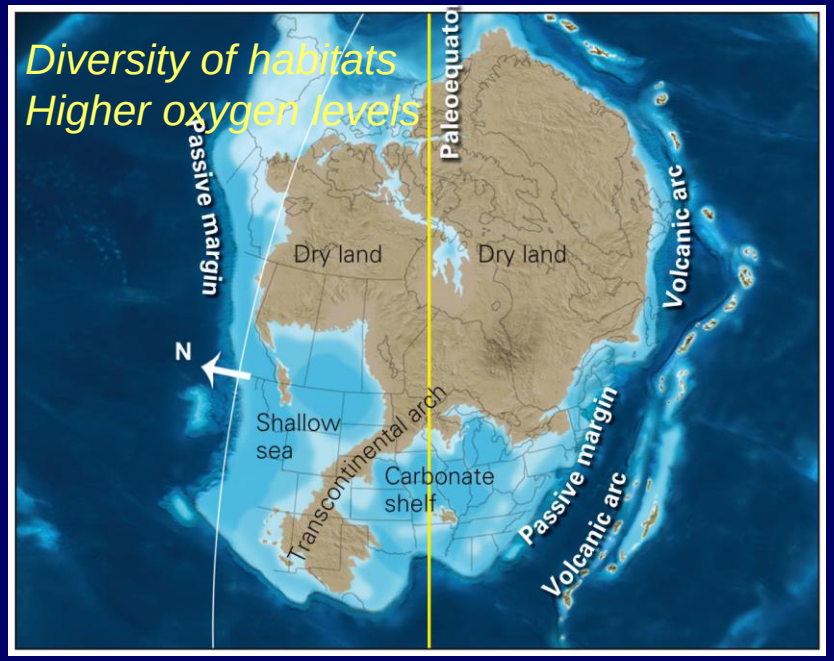


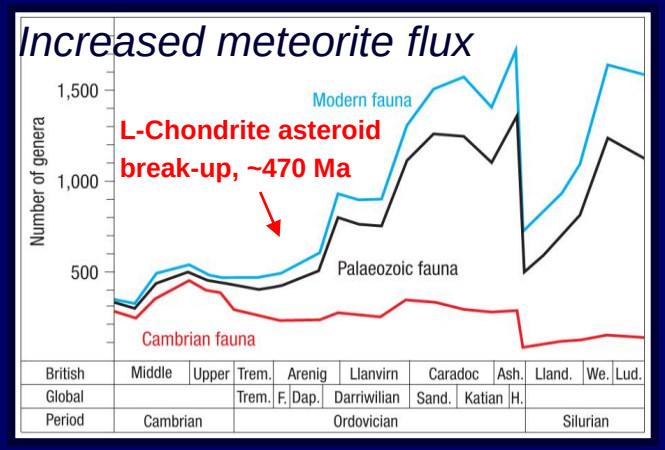
The Cambrian benthos dominated by deposit-feeders burrowing away were replaced by a Ordovician benthos of suspension-feeders living in communities with complex ecological structures.



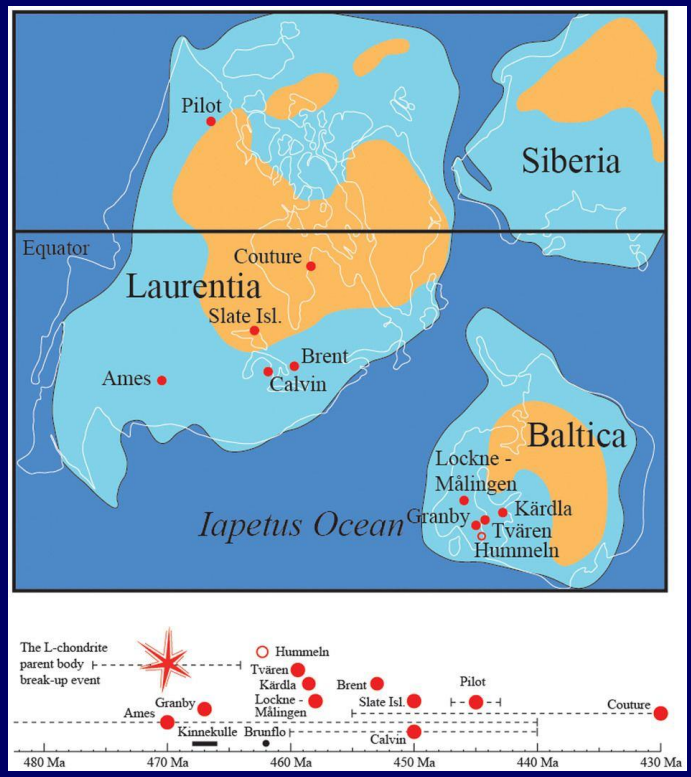
Location of mud mounds and mound complexes

Causes of GOBE





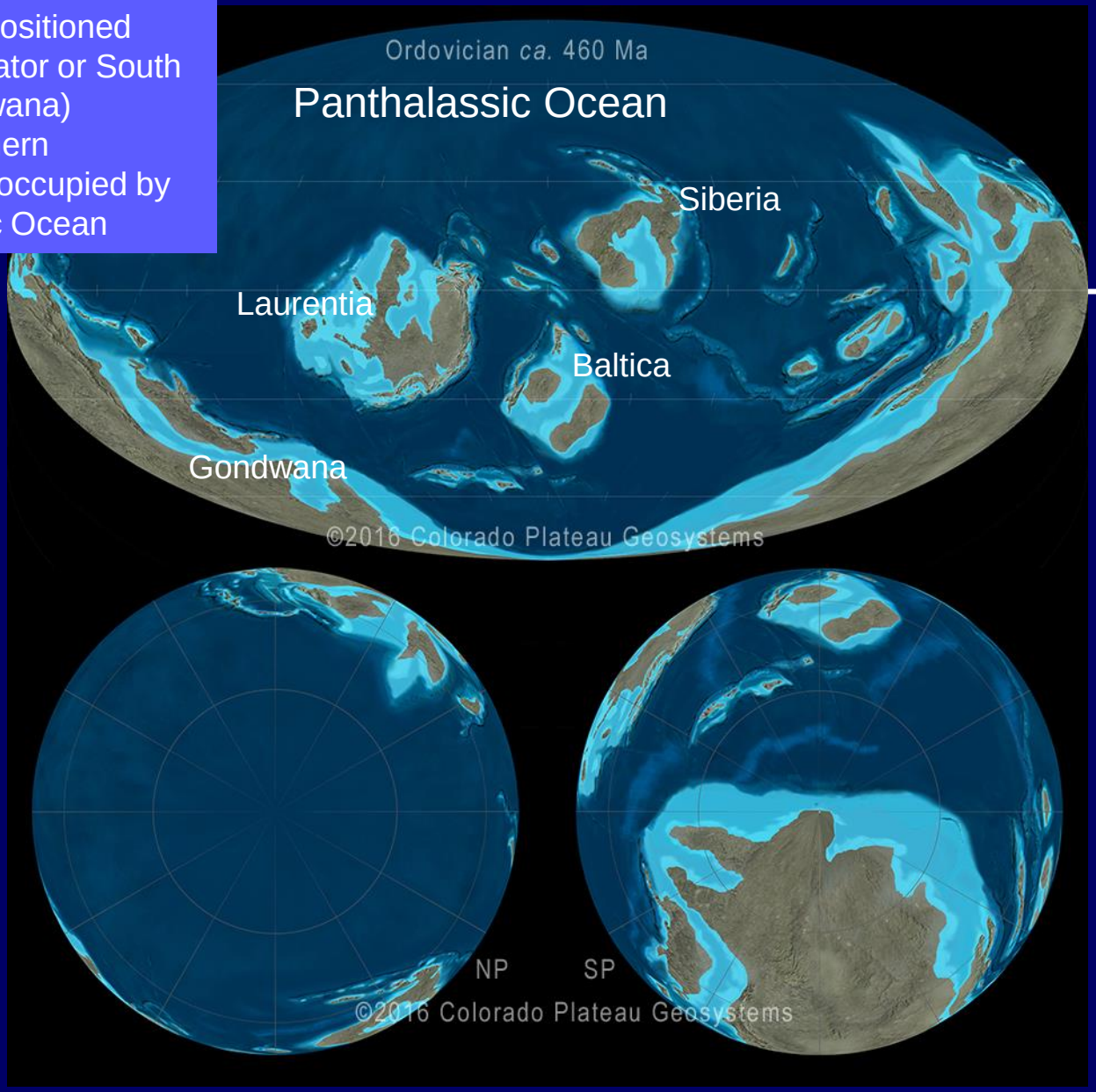
- Impact-related environmental changes may have accelerated diversity by creating a range of new habitat niches
- Frequent impacts may have destabilized ecological communities, allowing invasive species to take over and displace existing communities. e.g. a brachiopod takeover from trilobites in benthic communities,
- Development of suspension-feeding benthos with low metabolic rates better equipped to deal with and benefit from major environmental disruptions.



Ages of known impact craters potentially linked to the L-chondrite parent body breakup event.

Late Ordovician quasi-oceanic Northern Hemisphere

- Continents positioned close to equator or South Pole (Gondwana)
- 95% of northern hemisphere occupied by Panthalassic Ocean

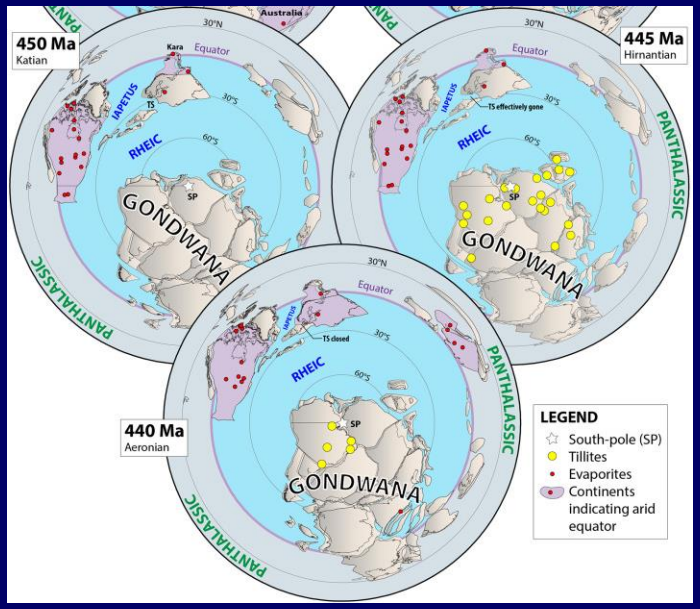


Equator

North pole

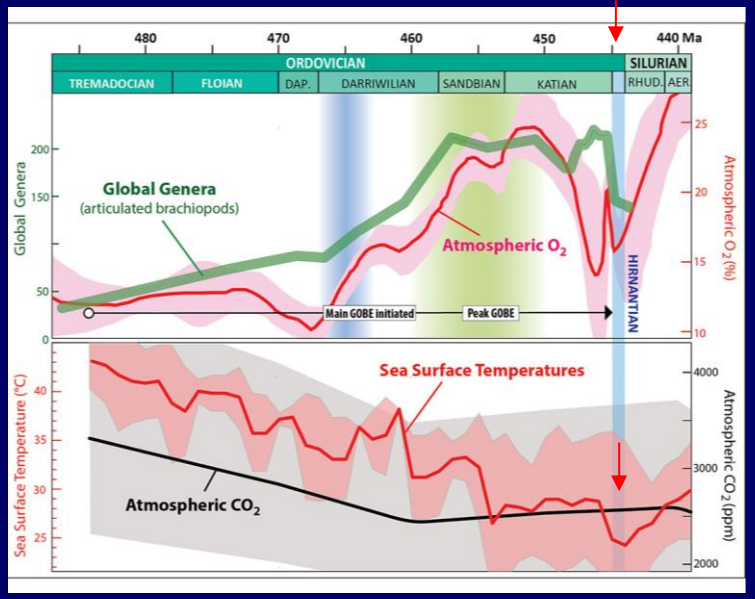
South pole

Late Ordovician Hirnantian glaciation

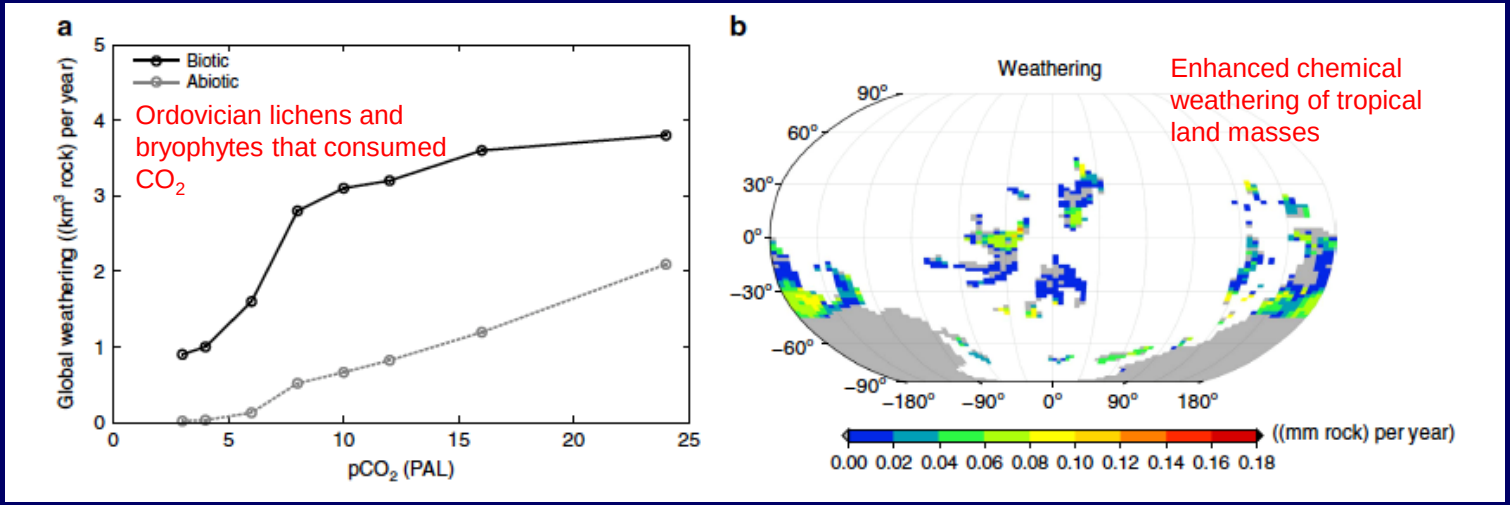


Distribution of glacial deposits on Gondwana

445 Ma, 0.5 Ma duration

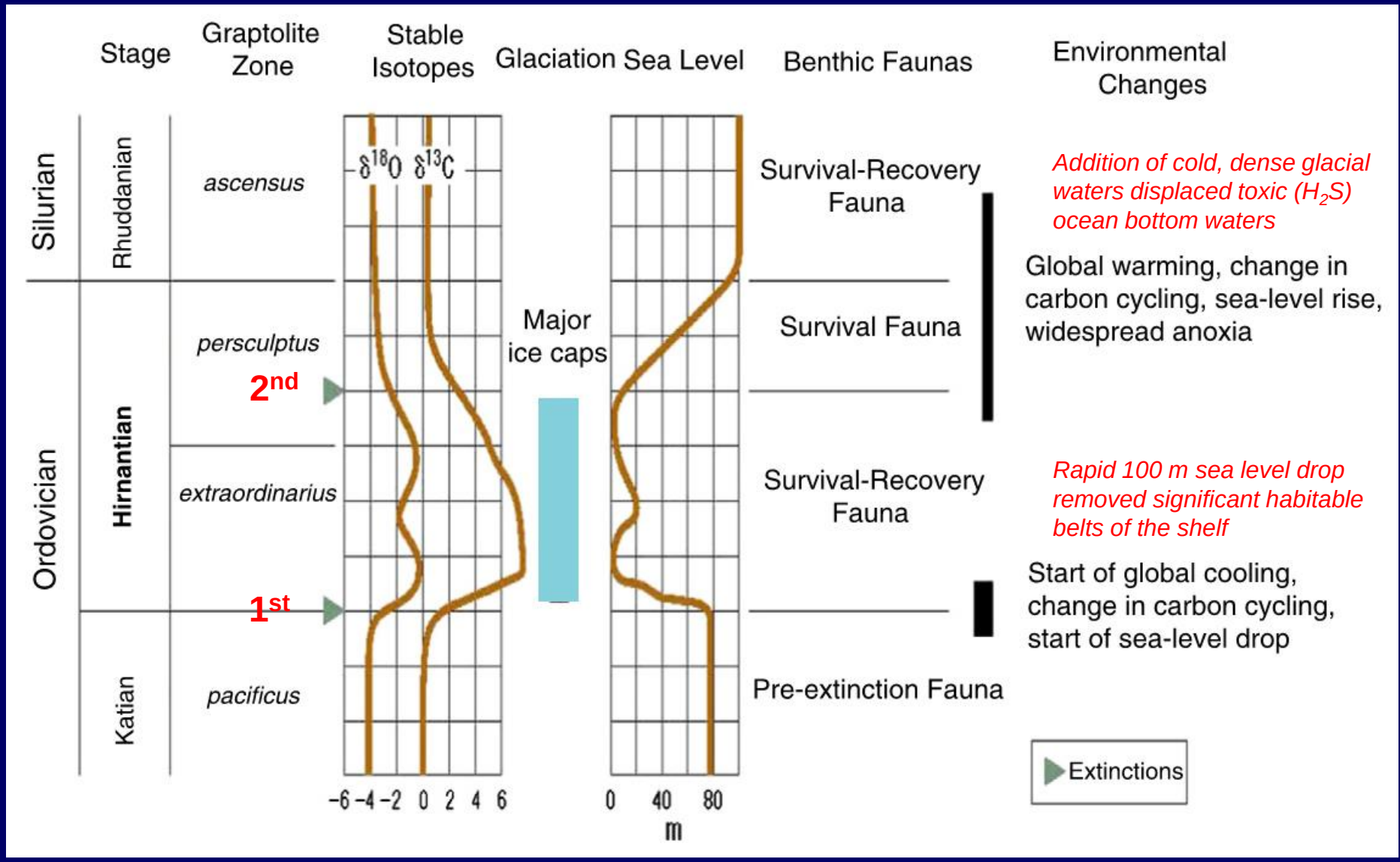


Decline in atmospheric CO_2 and SST

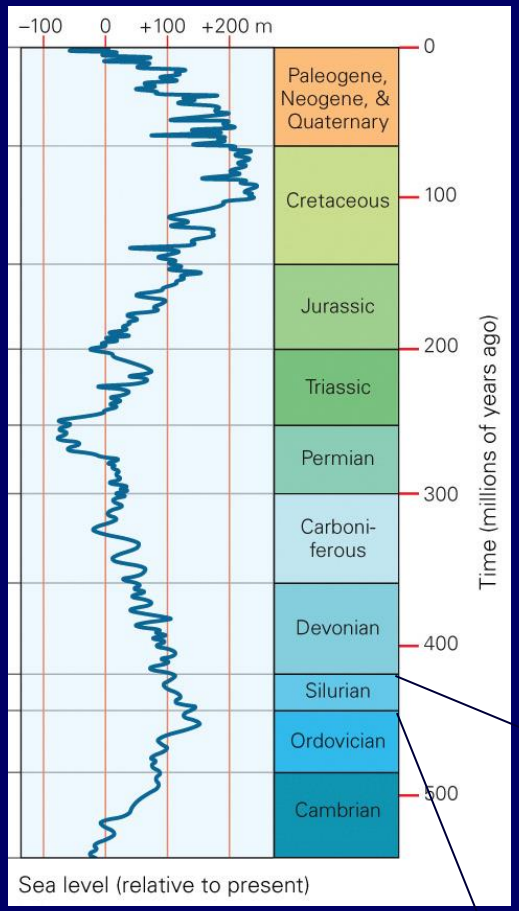


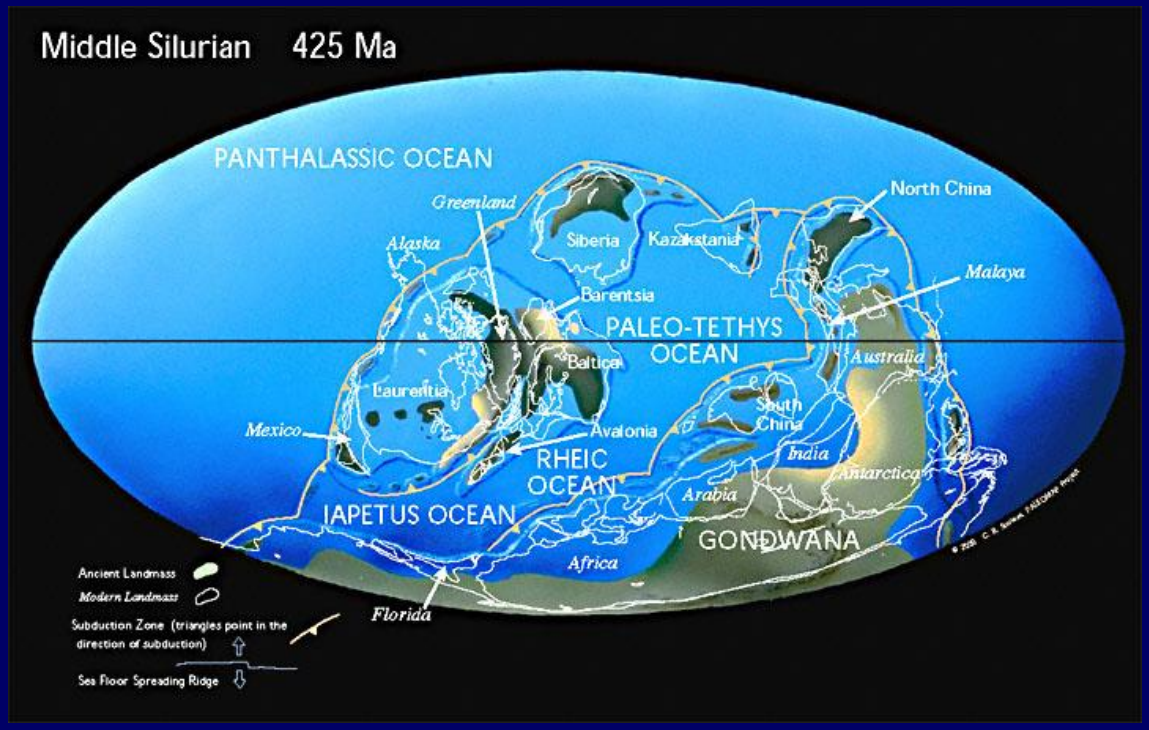
Weathering rates of siliciclastic rocks due to biotic versus abiotic mechanisms

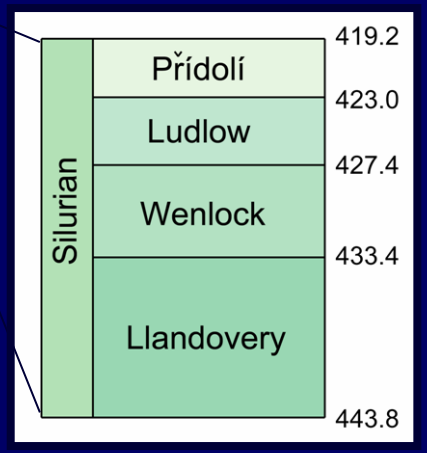
Late Ordovician glaciation and mass extinction event



Middle Paleozoic: Silurian (443.8 – 419.2 Ma)

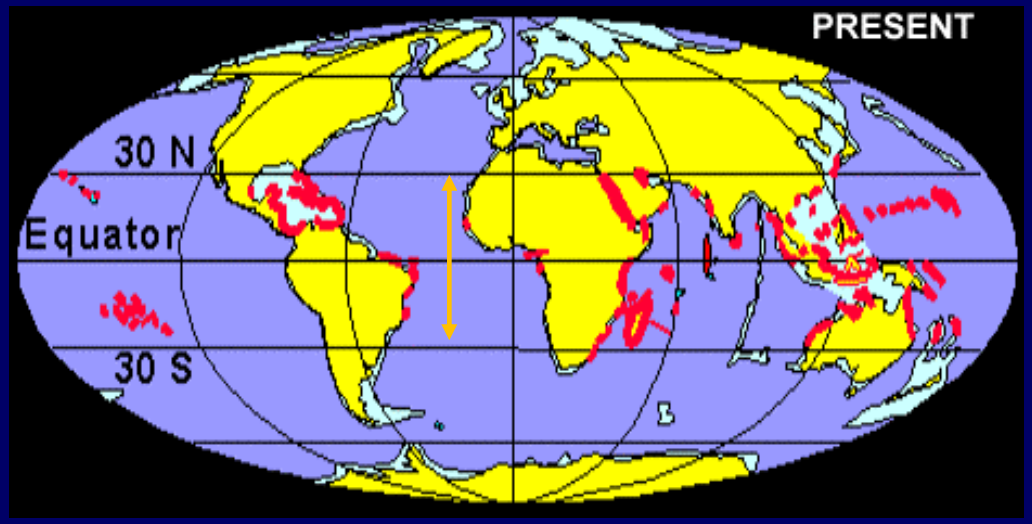






- Global climate warmed , sea level fell and Earth entered greenhouse phase
- Sustained recovery in biodiversity
- Evolution of terrestrial plants.
- First fish evolved and widespread reefs developed.

Changes in deposition of carbonate sediments over geological time

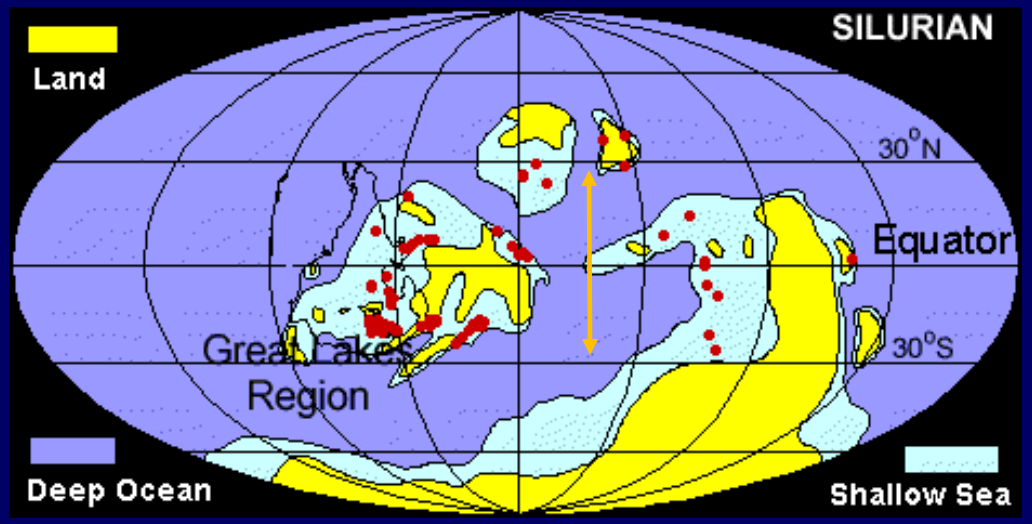


Type and abundance of carbonate reefs through time

Period	Relative abundance	Dominant skeletal elements of reefs
Neogene		Corals, algae
Palaeogene		
Cretaceous		Rudist bivalves, corals, stromatoporoids
Jurassic		Corals, sponges, stromatoporoids
Triassic		Corals, stromatoporoids
Permian		Algae, sponges, corals
Carboniferous		
Devonian		Corals, stromatoporoids
Silurian		
Ordovician		Corals, bryozoa, stromatoporoids
Cambrian		
Precambrian		Stromatolites

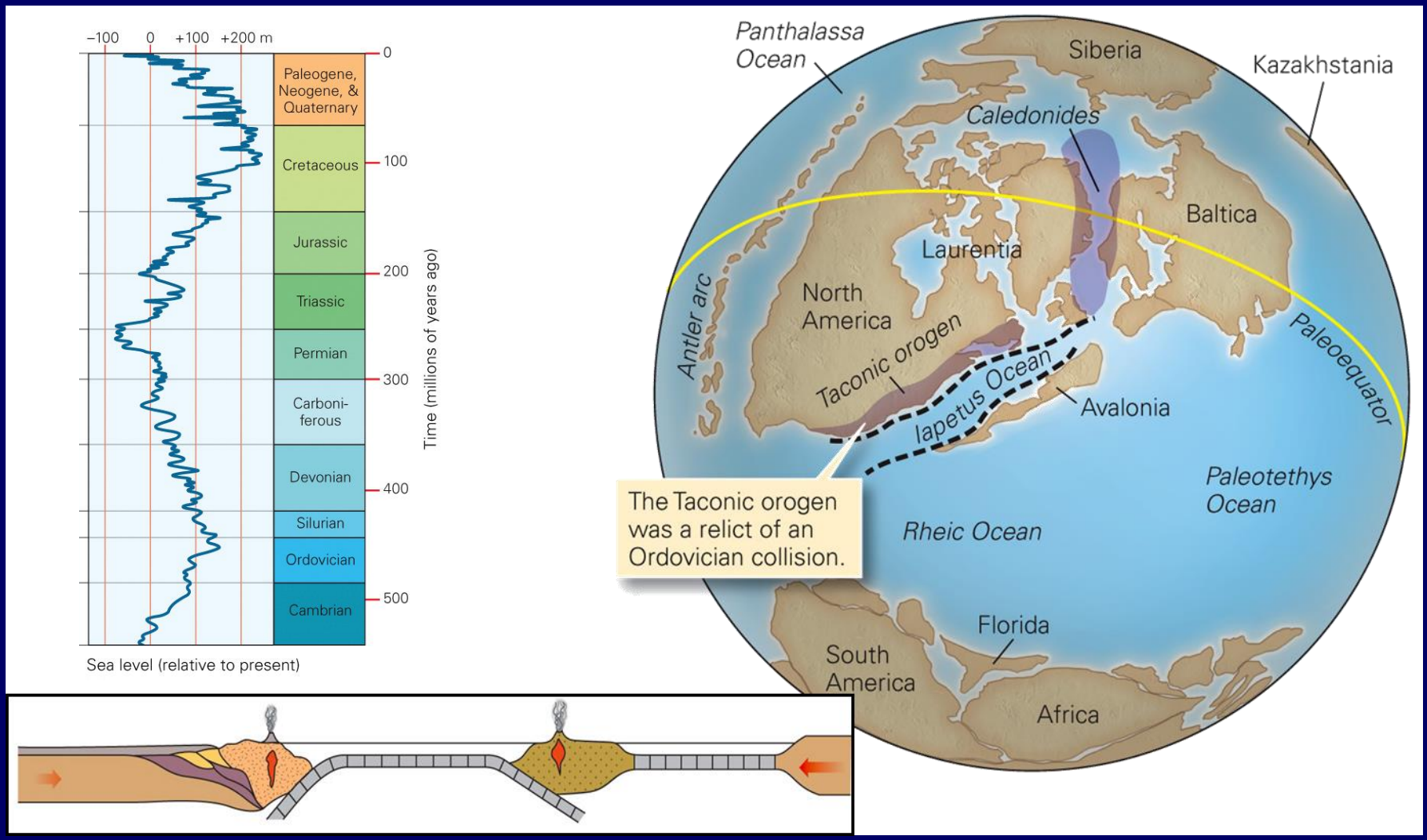


A Silurian reef comprised of 13 major groups of reef organisms and 57 species



Extensive reefs with diverse fauna

Continental movement (Silurian-Devonian) 444-359 Ma



- Collision of Laurentia, Baltica and Siberia = Laurasia
- Several orogenys took place → mountain building (Calendonian Orogen)
- Falling sea level → erosion of emerging continents → deposition of thick redbeds

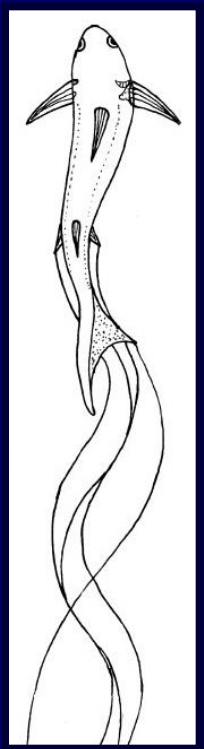
Devonian (419.2 - 358.9 Ma) Old Red Sandstones



Channel-floodplain cycles: red oxidised mudstones, calcretes, channel fills

→ ephemeral flow, semi-arid environment

Trace fossils as palaeoenvironment indicators





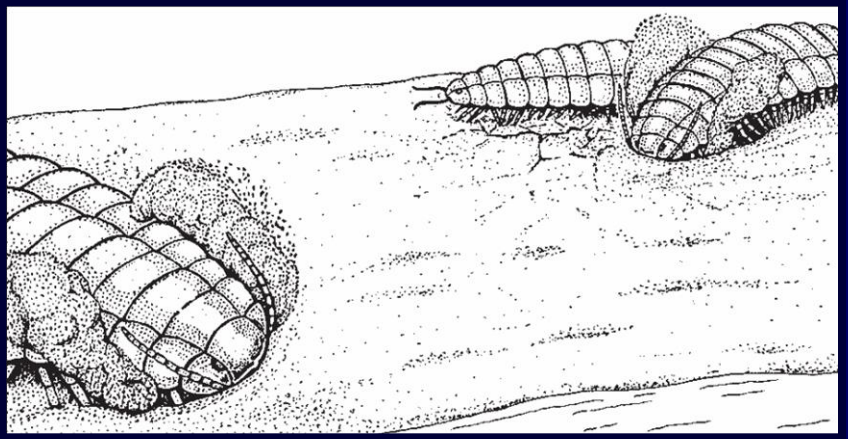
Undichna



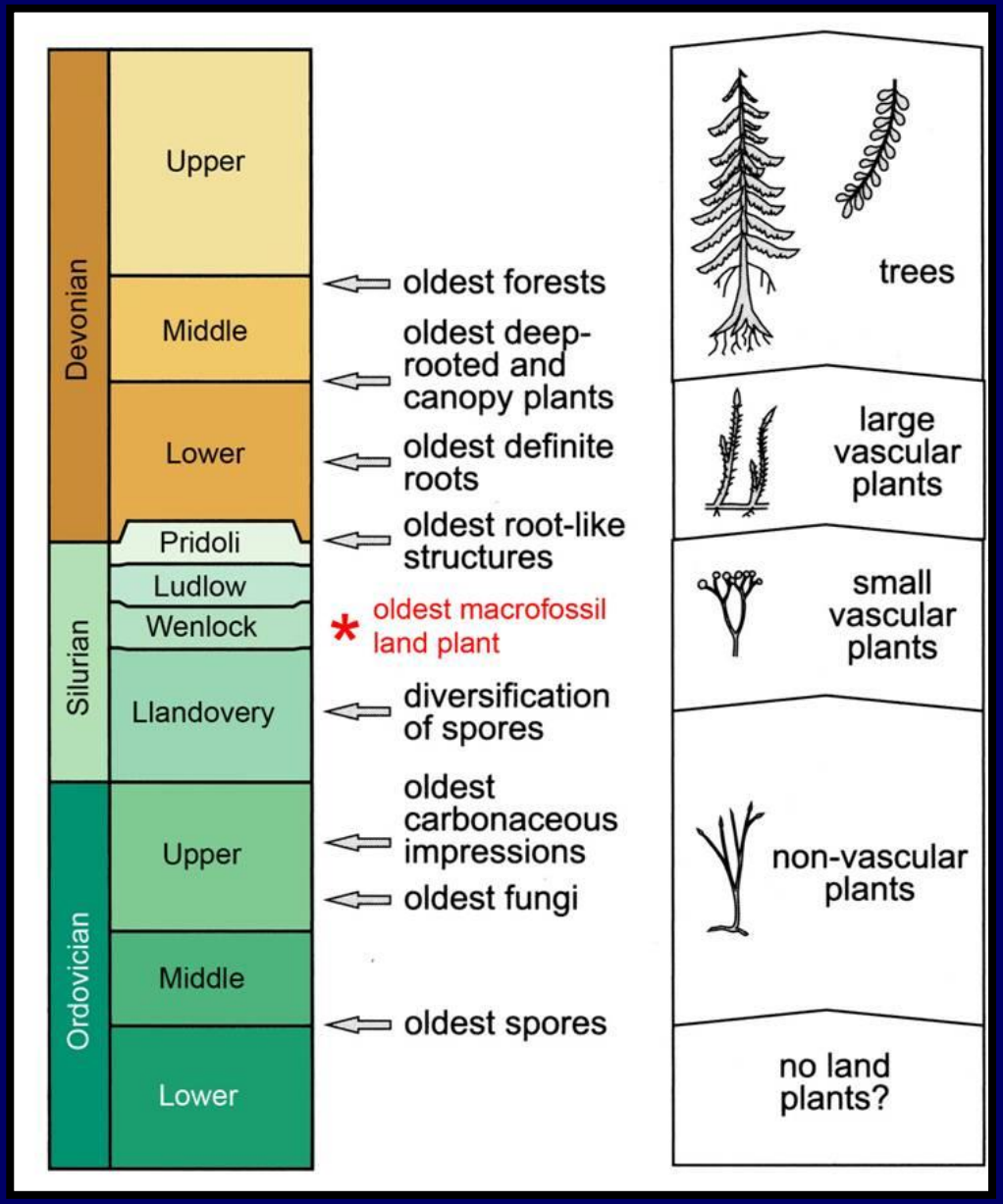
Beaconites



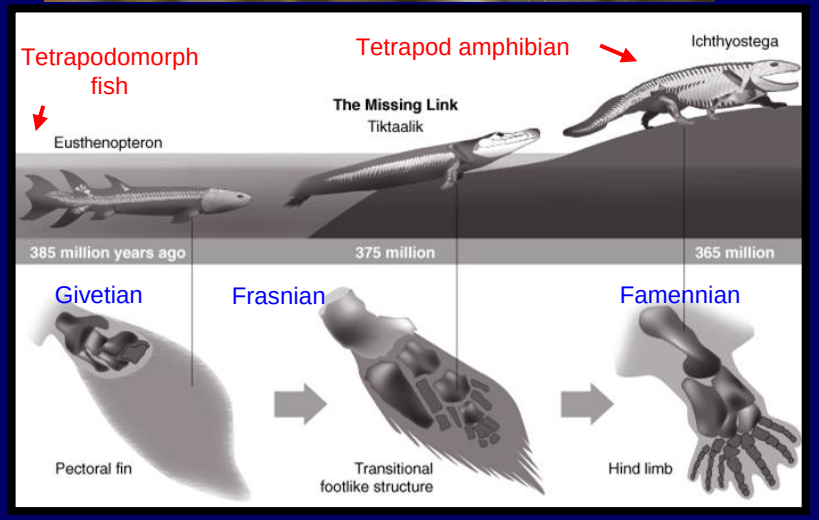
Diplichnites



Emergence of plants / colonization of the land

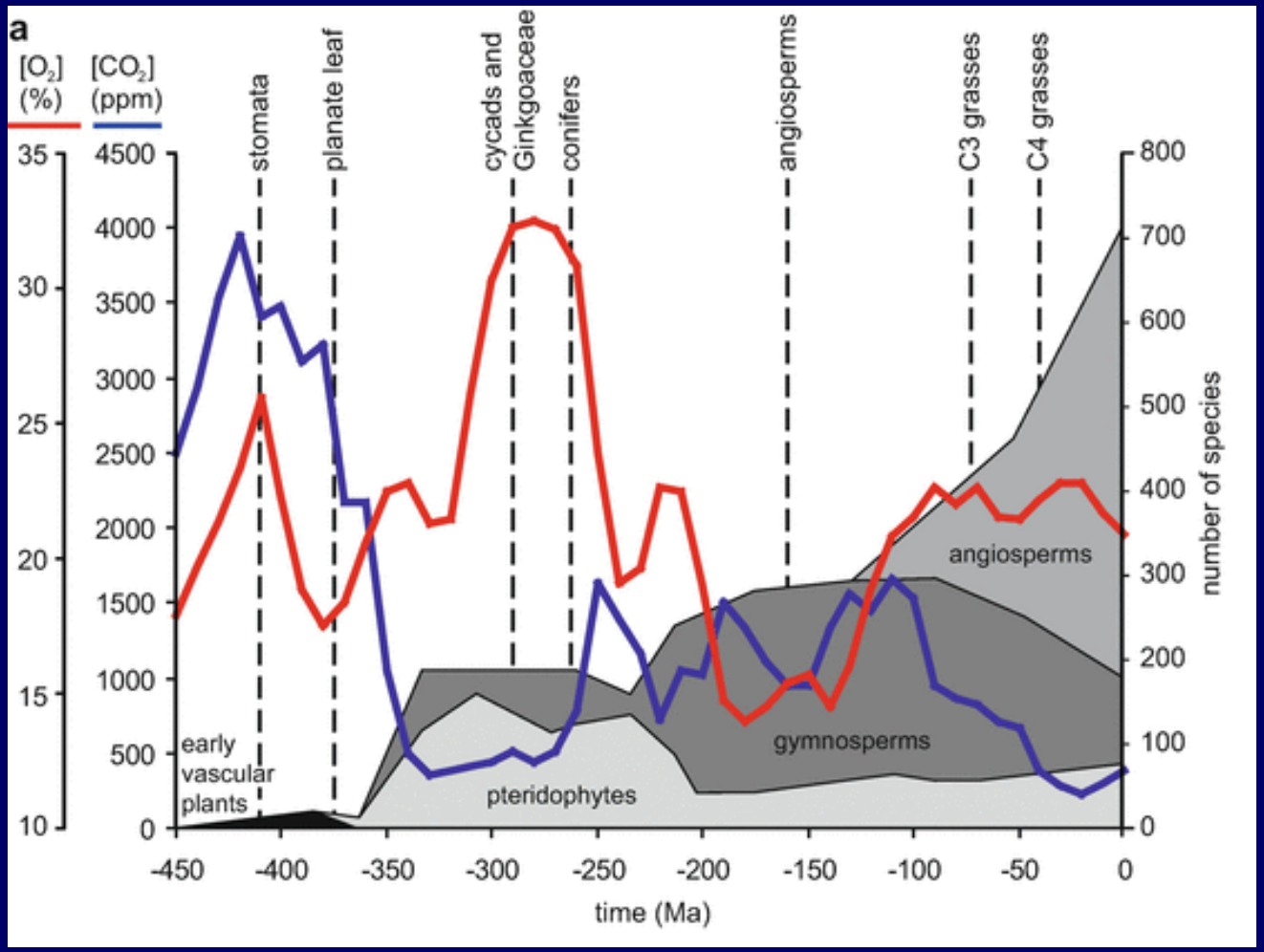






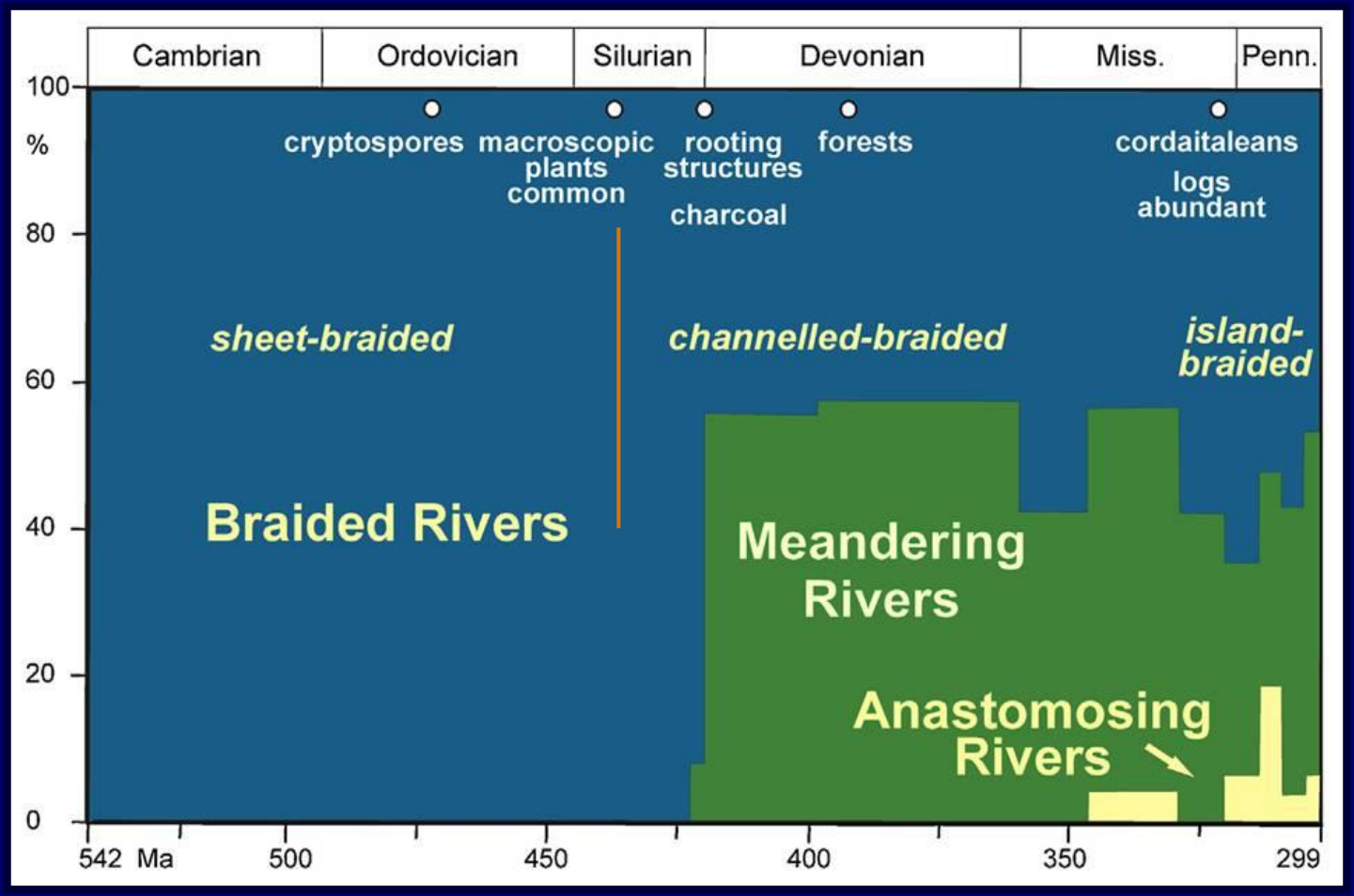
- vegetation provided food and shelter for animal life
- appearance of land arthropods at water margins from late Silurian
- evolution of air-breathing fish → appearance of the first tetrapods (amphibians) by the Mid-Devonian → appearance of the amniotic egg and reptiles by the mid-Carboniferous

Vegetation impact on the atmosphere and climate



Increased O_2 , lowered CO_2 → global cooling

Evolution of root systems stabilized flood plains



Abrupt appearance of meandering rivers in rock record at same time as plants began to colonize the land

→ change in the dominant channel planform from braided to meandering driven by plants?

Classification of rivers by their planform

The major control on channel planform is the ‘strength’ of the banks



Rangitata River, Canterbury Plains, NZ

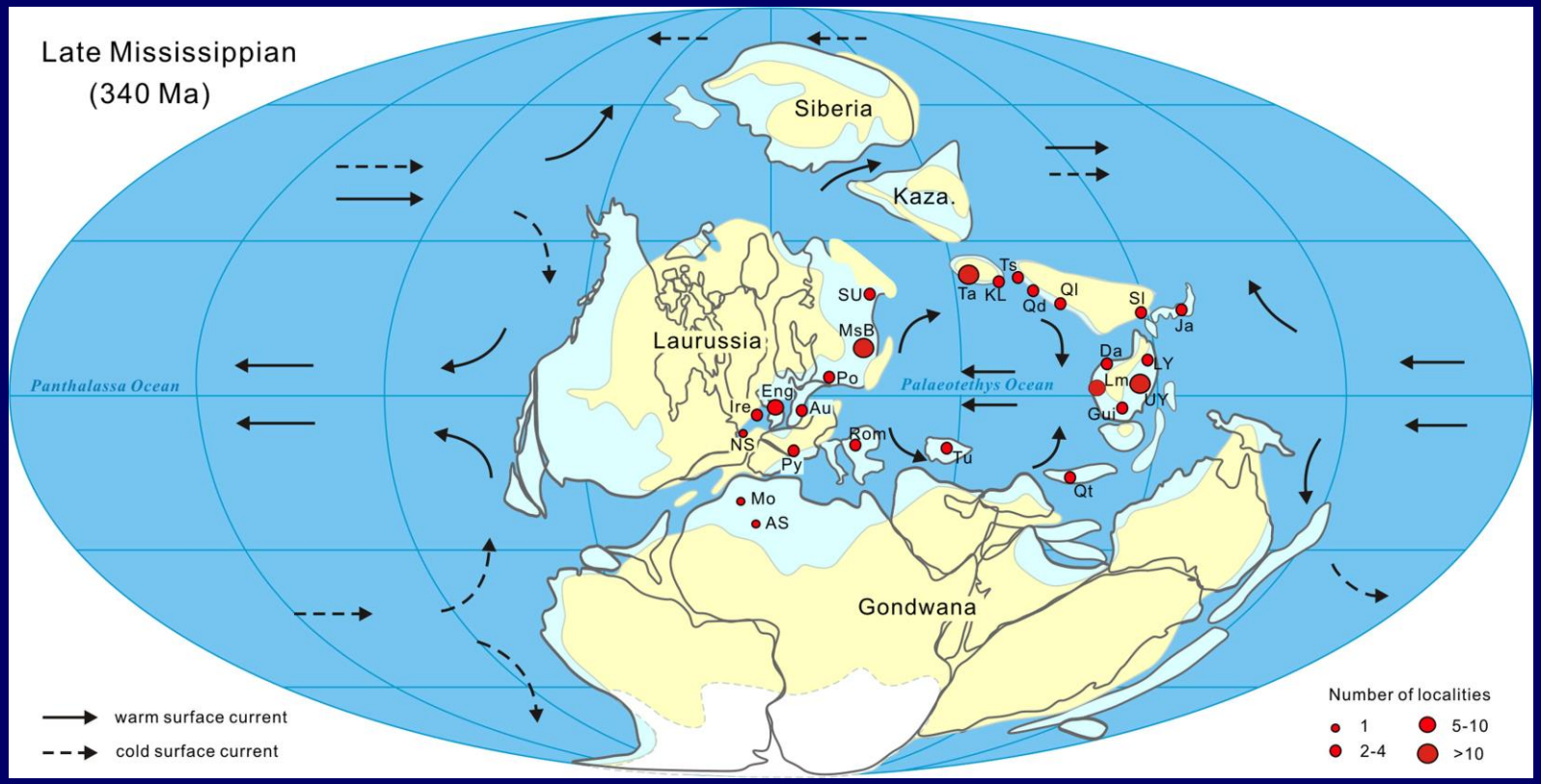


Río Socopo, Venezuelan Andes

Today, land plants make channel banks ‘cohesive’ so a meandering planform can develop

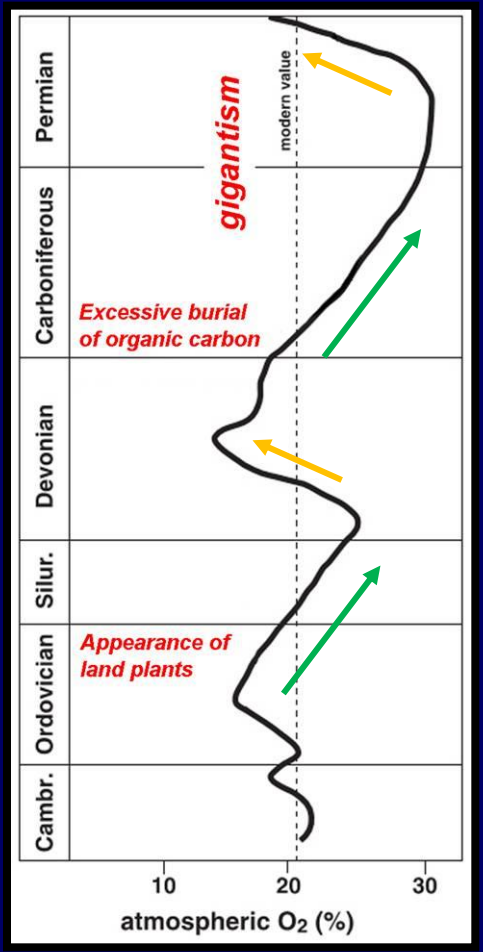
Therefore, prior to land plants were rivers only braided?

Late Paleozoic: Carboniferous (358.9 - 298.9 Ma)



- Climate cooled in late Paleozoic
- Seas regressed from continents
- Continental land mass Laurussia assembled at equatorial and tropical latitudes with swamps forming thick coal deposits
- Gondwana and Siberia were at higher latitudes and by Permian were partially covered by ice sheets

Oxygenation of the atmosphere




Meganeura, dragonfly with 70 cm wingspan


Arthropleura

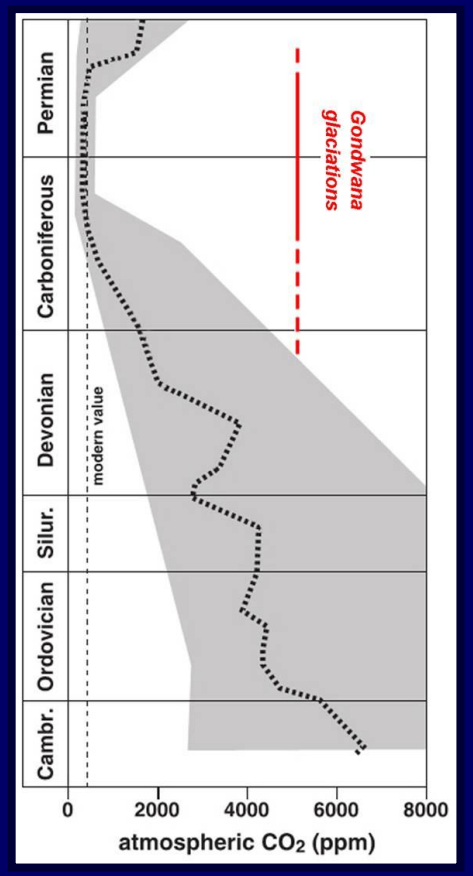
Increase in atmospheric O_2 levels

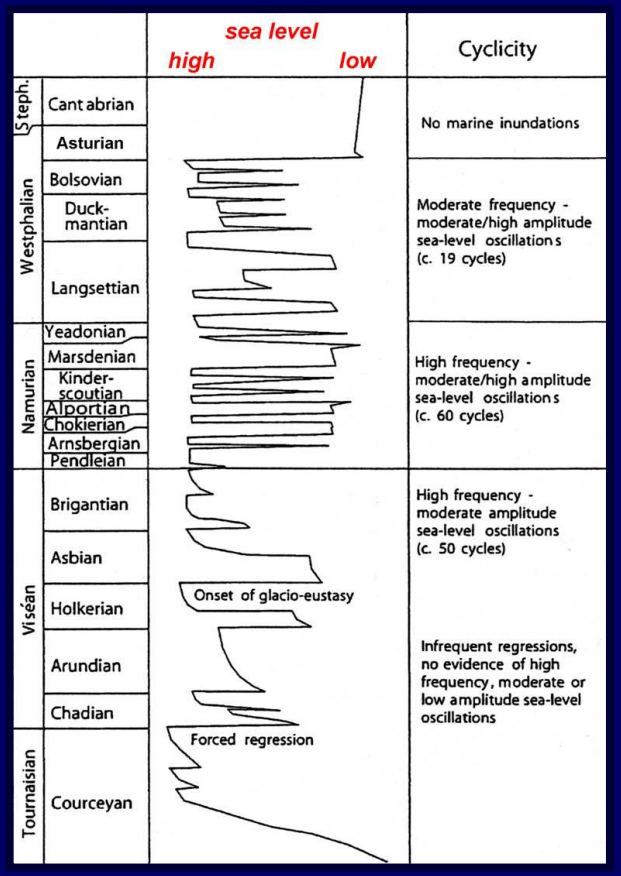
- Ord-Sil = appearance of land plants (photosynthesis)
- Dev-Carb = burial of organic matter, evolution of decay resistant lignin, proliferation of forests and swamps.

Decreases in O_2 levels (early Dev and end Perm)

- Changes in land vegetation related to climate change and latitude of major continents

Climate and sea level fluctuations





- Carboniferous stratigraphy shows strong cyclicity in response to sea level changes resulting from repeated growth and melting of polar ice sheets on Gondwana.
- Note this ice-age coincides with low atmospheric CO_2 levels similar to modern.
- The most extensive pulses of glaciation began in the Asbian and continued into the Mid-Permian
- The periodicity of sea level fluctuations correlates with the ~ 100 ka Milanković orbital eccentricity cycle
- Continents at tropical latitudes experienced the glaciations indirectly through sea level changes – they were not glaciated themselves.

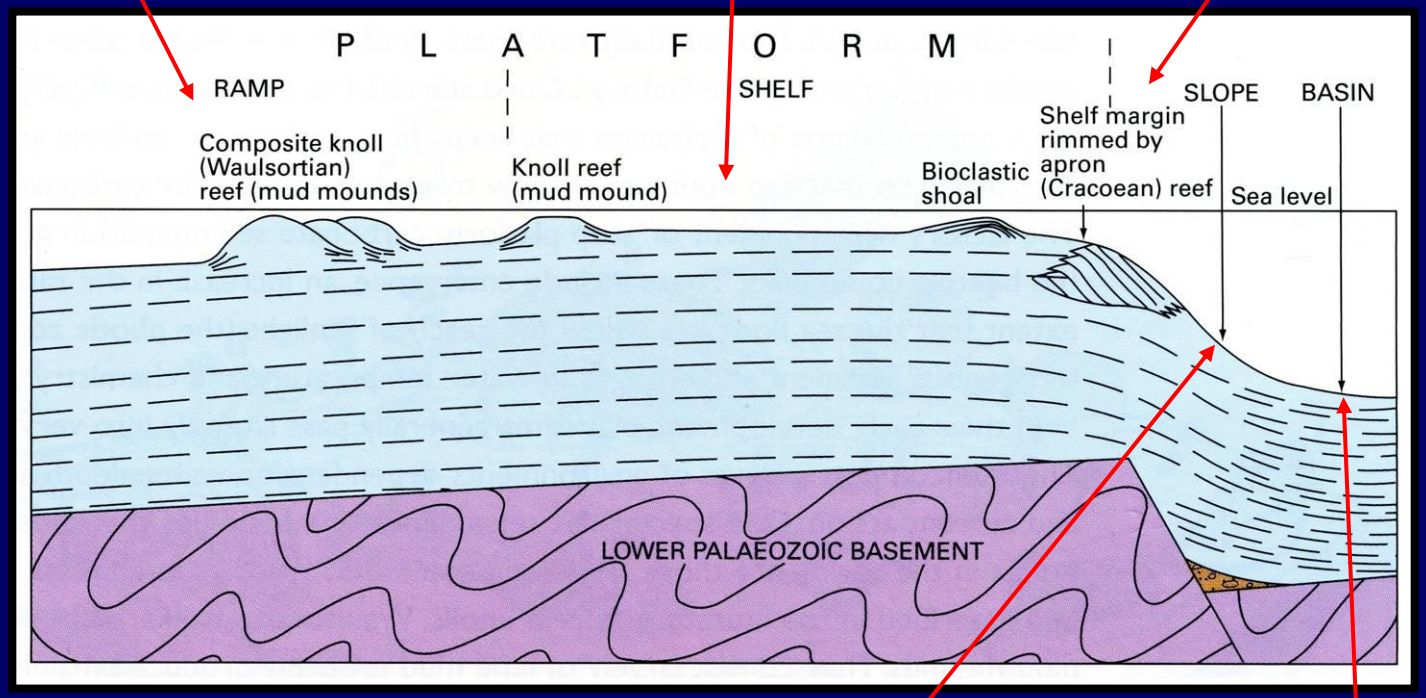
Carbonifierous transgression / carbonate sedimentation

At equatorial latitudes when siliciclastic input was not too high and sea level rise was not too fast, sedimentation in the tropics was dominated by carbonates

Thinly bedded, bioclastic limestones with cherts and micritic limestones forming mud mounds

Thickly bedded, massive, bioclastic limestones

Framework community constructed by an encrusting biota of microbes, sponges, bryozoans, and corals



Limestone conglomerates and turbidites with soft-sediment deformation features

Thinly bedded, cherty, fine-grained limestones and dark mudstones

Carboniferous limestone quarries



Coldstones Quarry, Greenhow, near Pateley Bridge, Yorkshire

Tunstead Quarry, near Buxton, Derbyshire



Late Carboniferous coal measure formation

Coal is of vegetable origin

- >2/3 rock is solid organic matter
- contains visible plant remains
- all transitions from peat soil to coal seam





Peat



Anthracite



Transition with temperature is:
Peat > Lignite > Bituminous Coal > Anthracite

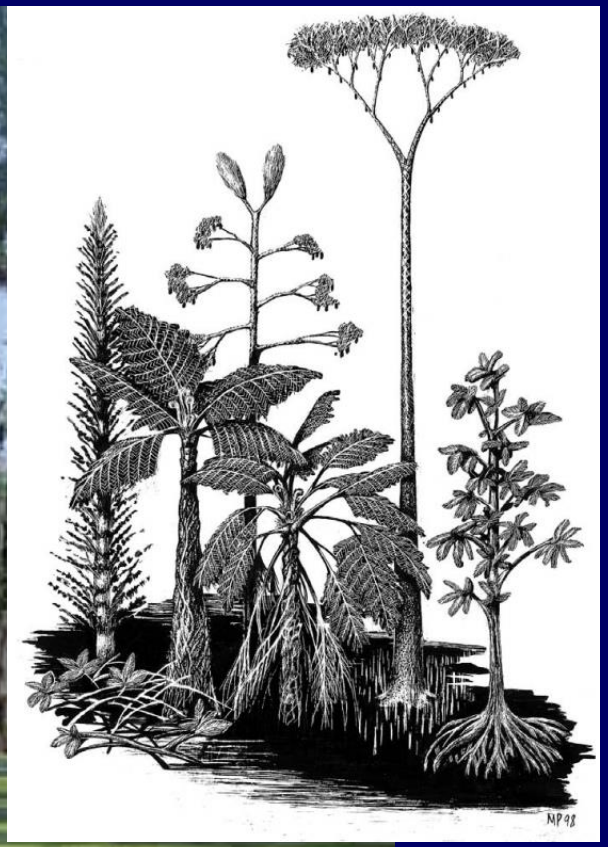
Late Carboniferous coal measure formation



- Plant matter accumulates rapidly in stagnant waters e.g. swamps or lakes on a river floodplain
- Oxygen is quickly used up in the oxidation process
- The water becomes anoxic and partly decomposed plant material remains
- Swamps are periodically inundated by sediment during times of flood of the rivers
- In lakes starved of large amounts of sediment input, organic-rich muds accumulate
- Burial and compression of sediment and plant material expels water, forming peat
- As burial continues the temperature and pressure increases and the ‘grade ‘ of the coal increases

Modern analogue?



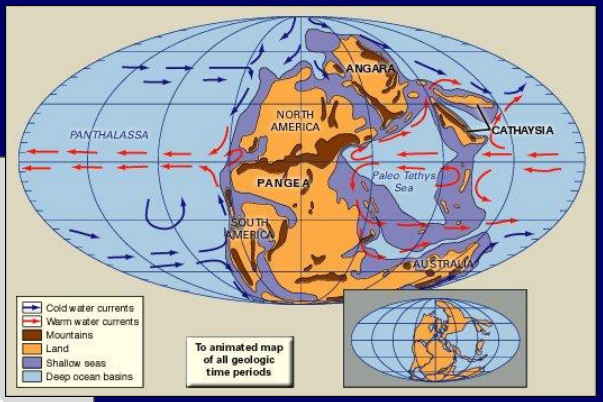


Reconstructed growth habits of mature, Carboniferous coal-swamp plants, max height 30 m.

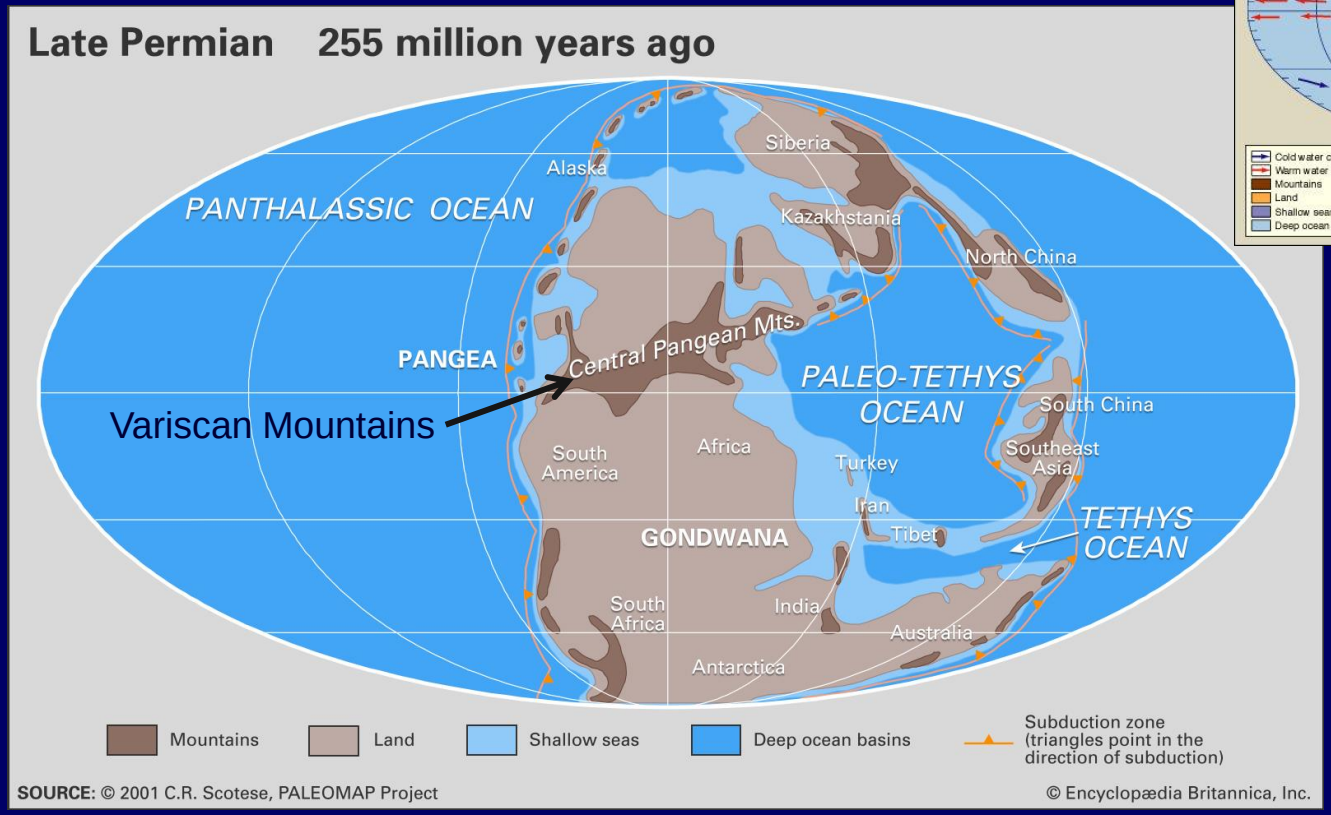
A swamp such as the Everglades in Florida is probably the closest modern equivalent to ancient coal–forming environments

Pangea: only single supercontinent of Phanerozoic

The collision of the two supercontinents, Laurussia and Gondwana in the late Carboniferous (Variscan Orogeny aka Hercynian orogeny) created the single supercontinent Pangea



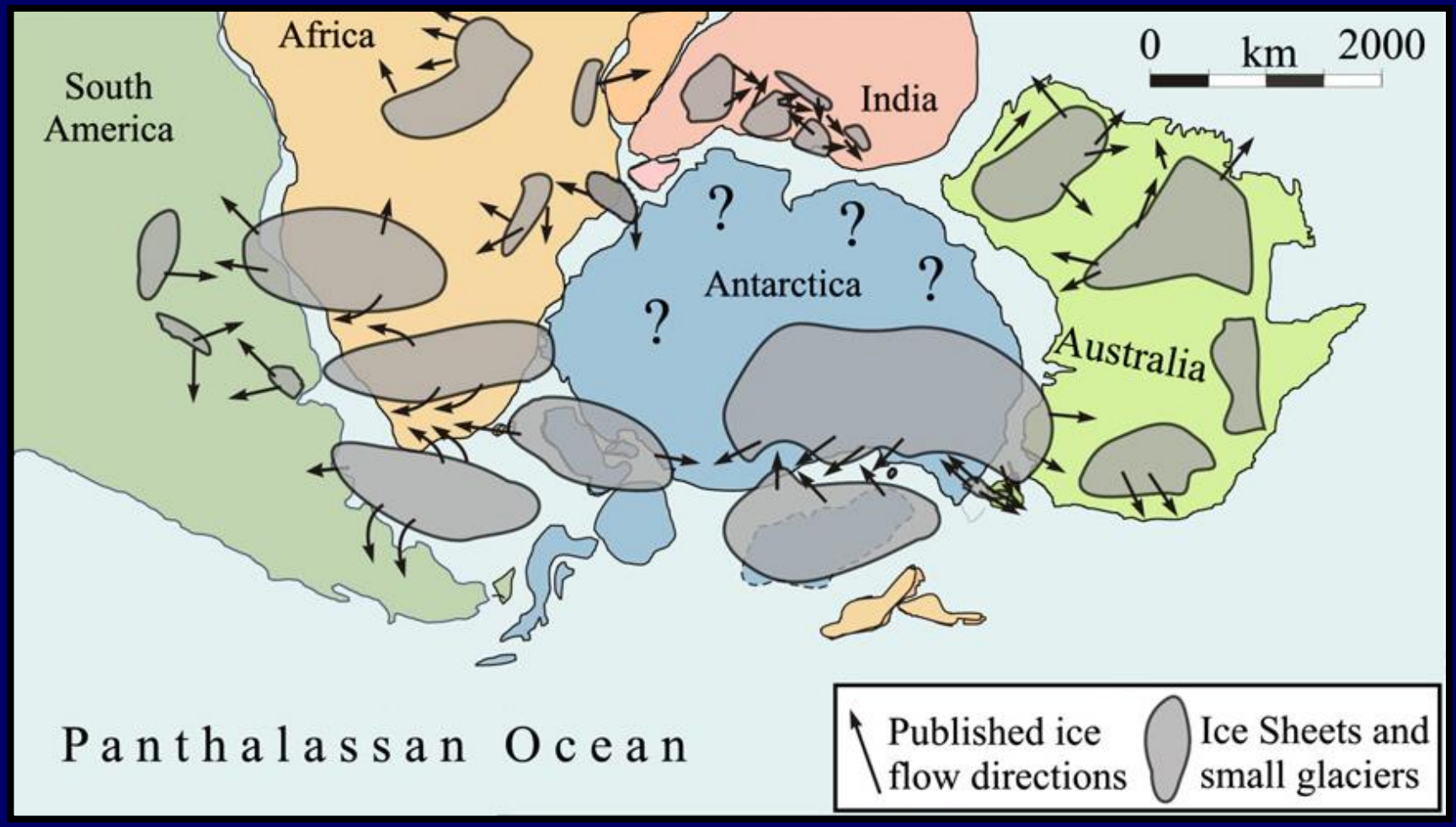
Effect on ocean circulation



The Variscan gets its name from the Germanic Varisci tribe

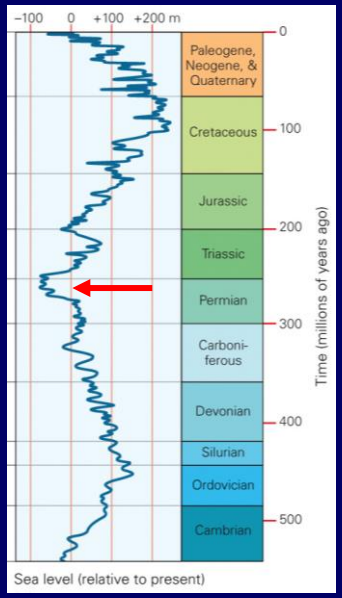
Pangaea straddled the equator and extended toward the poles

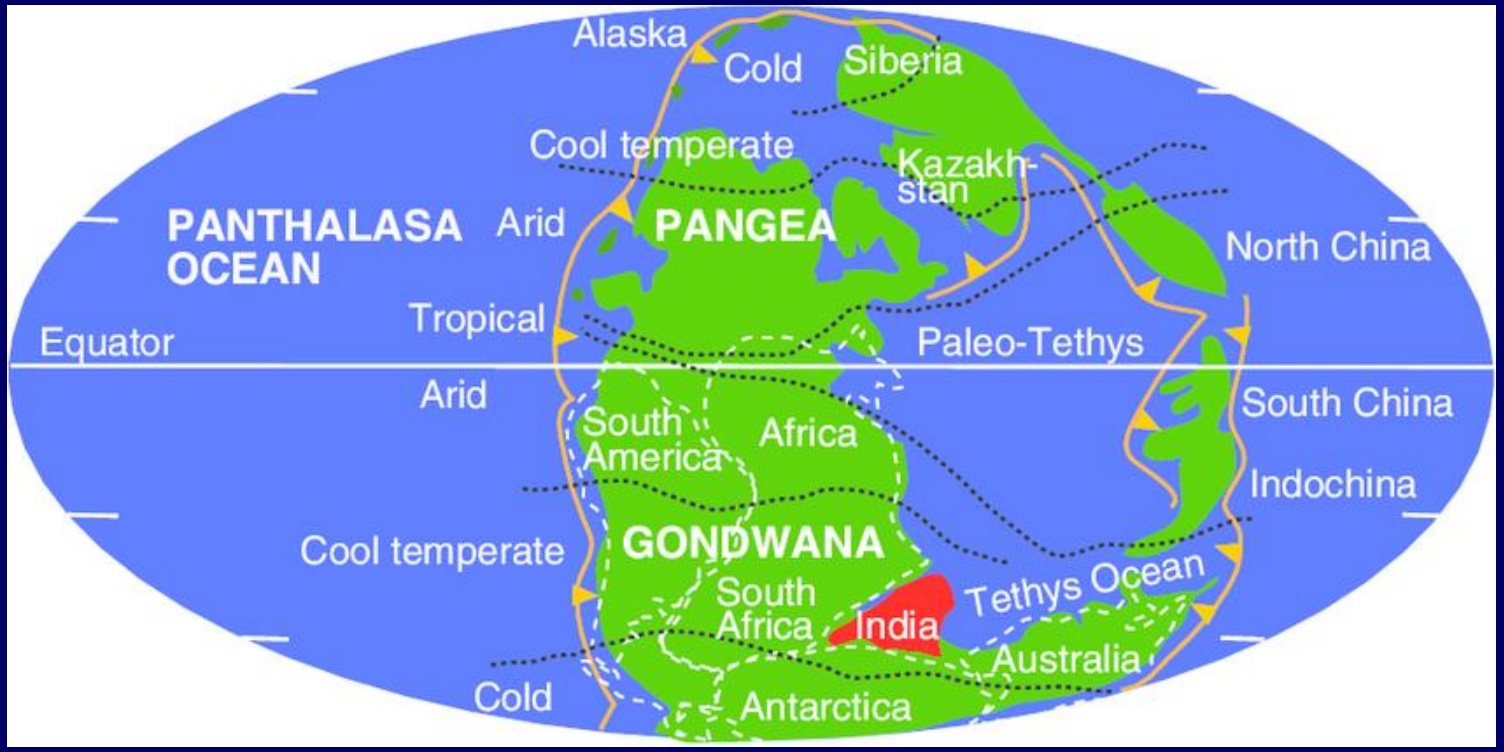
Early Permian environment: high latitudes



Reconstruction of the Gondwanan ice sheets during the deepest glacial conditions in the Early Permian.

Permian environment: low latitudes

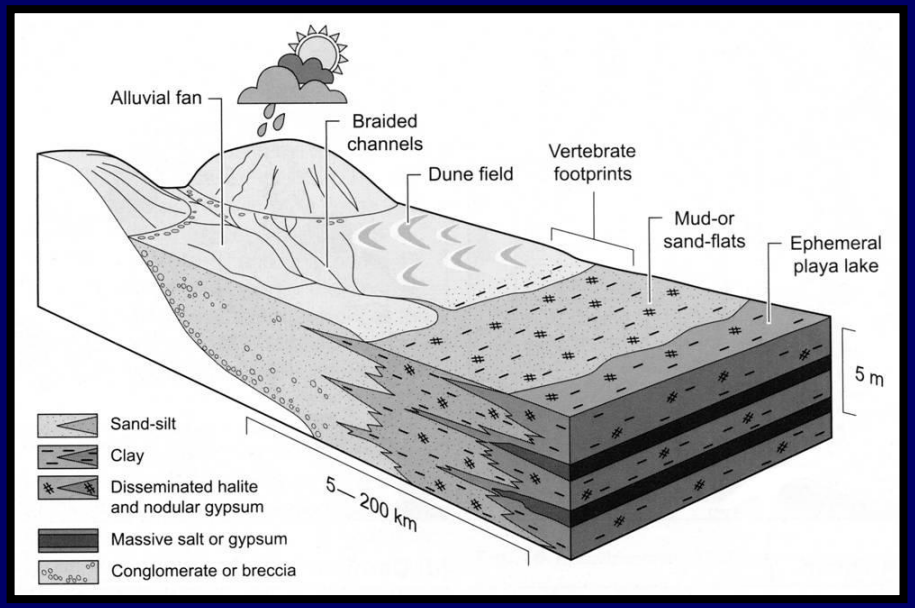




- Sea levels were at their lowest due amalgamation of continents forming supercontinent Pangea
- Arid and semi-arid climate at latitudes 0-45 degrees with extreme seasonality (huge continental interior- ‘continental climate’)
- Carboniferous plants (pteridophytes - spores) replaced by seed-bearing plants (gymnosperms)

The Permian deserts

The Carboniferous rainforest collapse left behind vast regions of desert within the continental interior



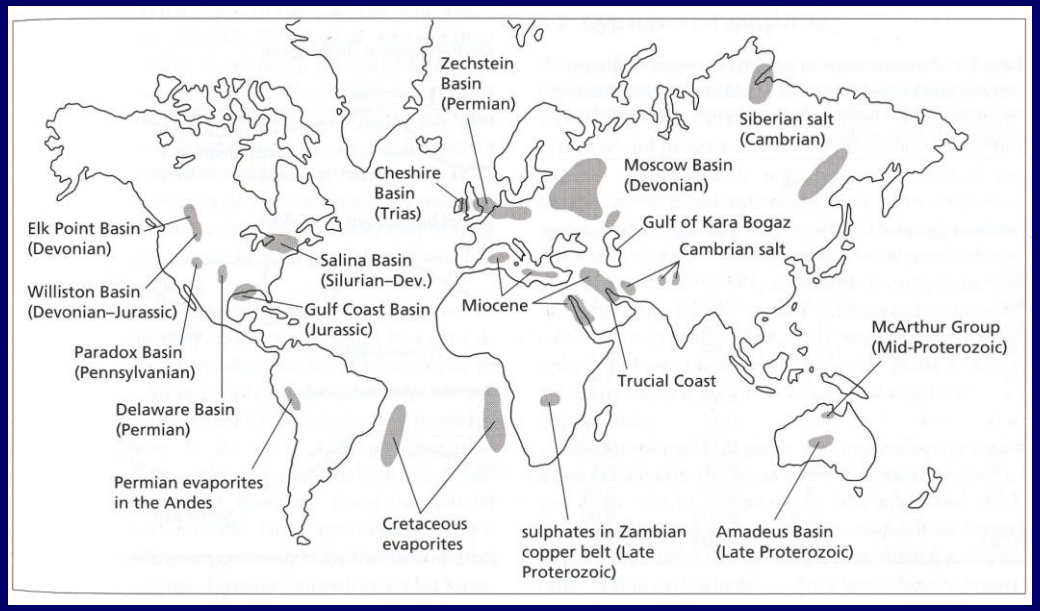
Basal conglomerates and pebbly sandstones deposited in ephemeral streams → aeolian dune fields → playa lake deposits

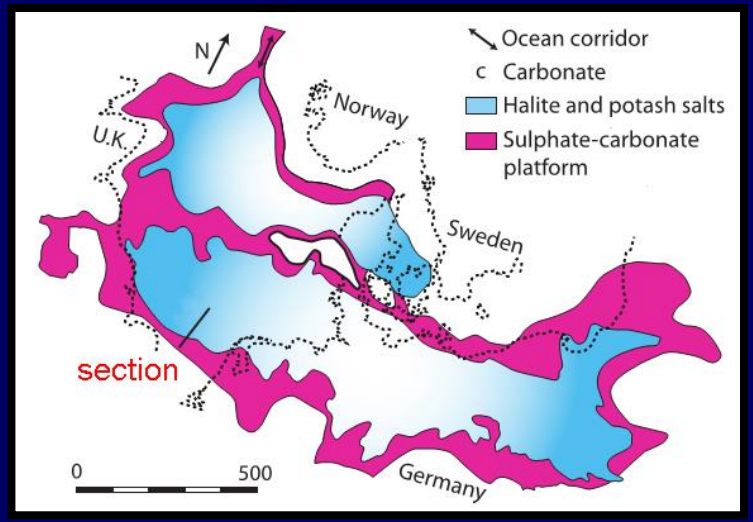


Top: Large scale aeolian dune bedding, Ballochmyle Quarry, Mauchline, Ayrshire.

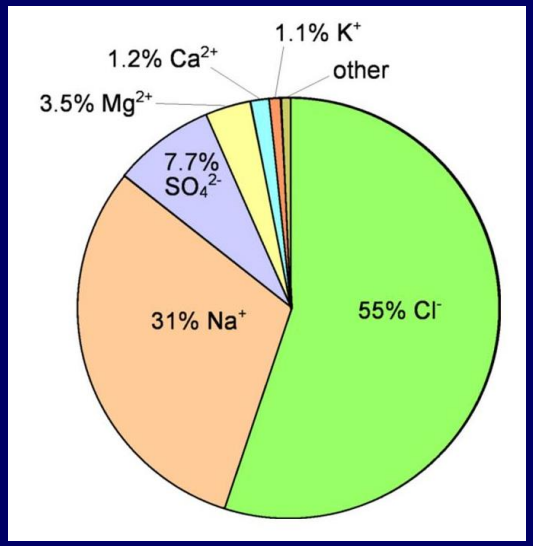


Zechstein Sea: Evaporite deposition





The late Permian Zechstein Basin, 1200 m thick evaporite deposits



Abundance of principal ions in seawater

Seawater normally contains 3.5% by weight of dissolved salts:

- Chloride
- Sodium
- Sulphate
- Magnesium
- Calcium
- Potassium

decrease weight %



Death Valley, California

Precipitation of evaporites

Evaporation of sea water precipitated:

increasing evaporation ↑

K- and Mg-salts

↑

Halite (NaCl)

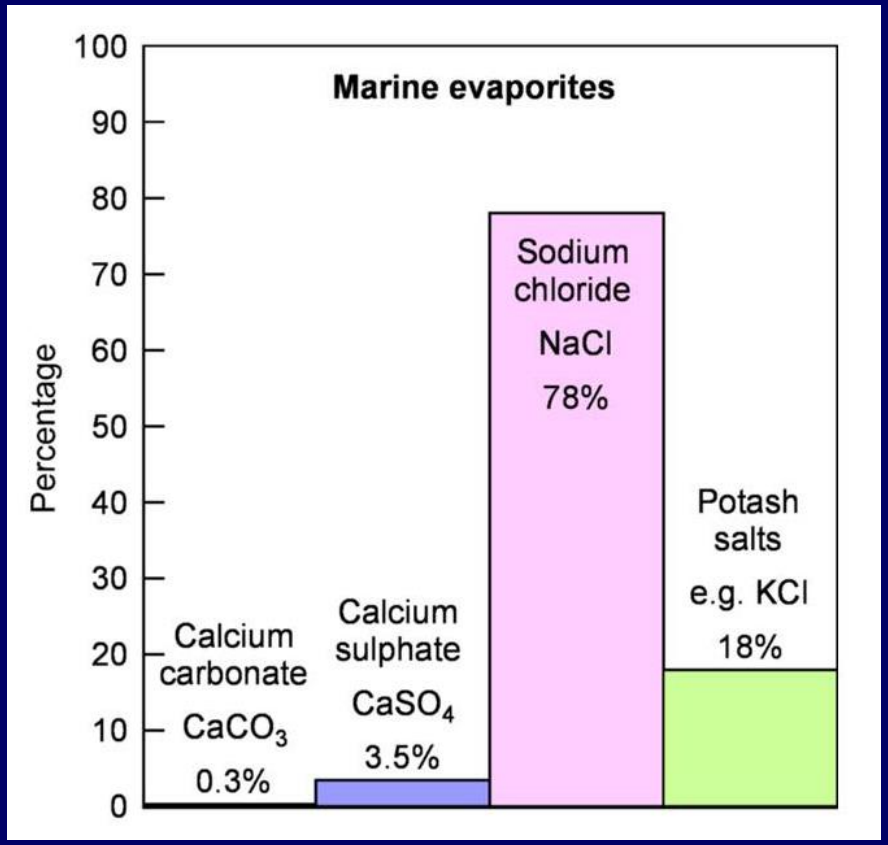
Gypsum ($CaSO_4.2H_2O$)

↑

carbonates

1000m deep seawater evaporated to dryness would leave ~16.5 m thickness of salts:

- Carbonate ~0.1 m
- Gypsum ~0.6 m
- Halite ~12.9 m
- Mg and K salts ~2.9 m



Proportions of minerals precipitated by evaporation of seawater

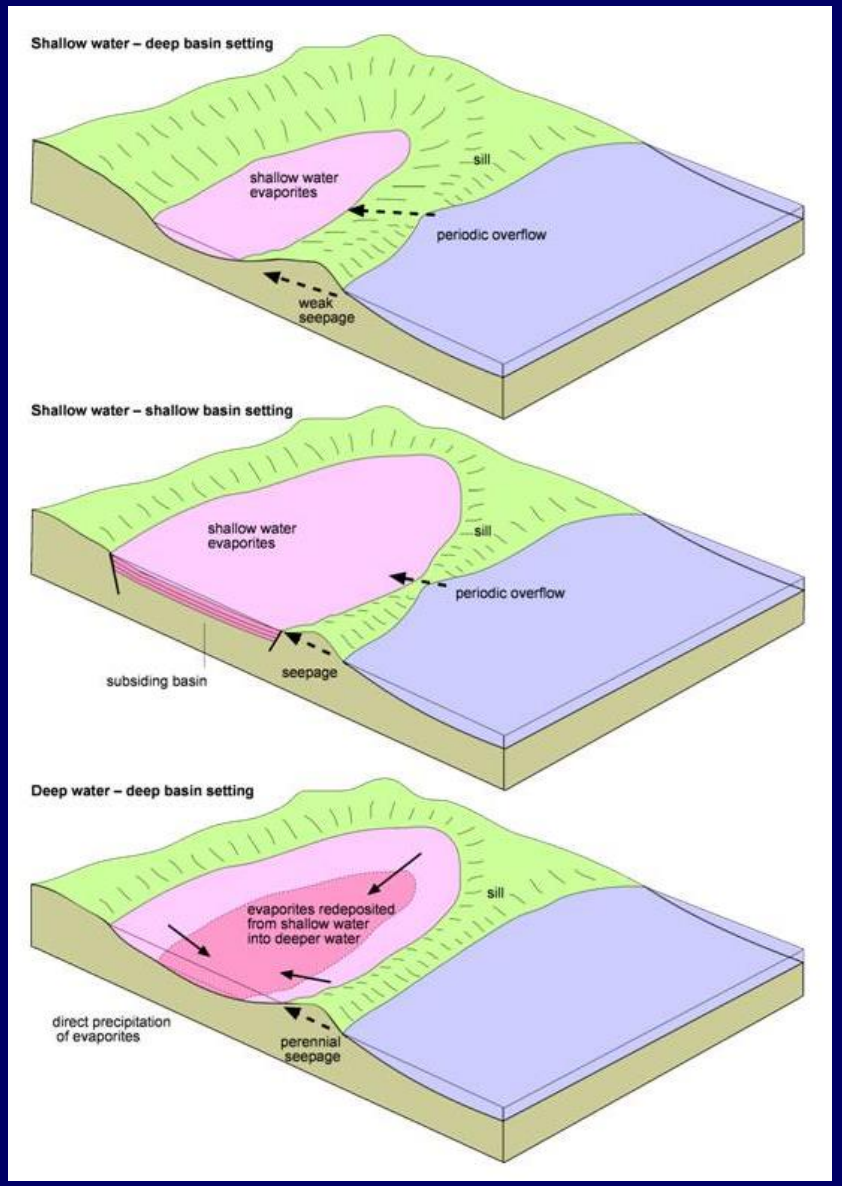
“Barred basin” depositional models

“Evaporating dish” process

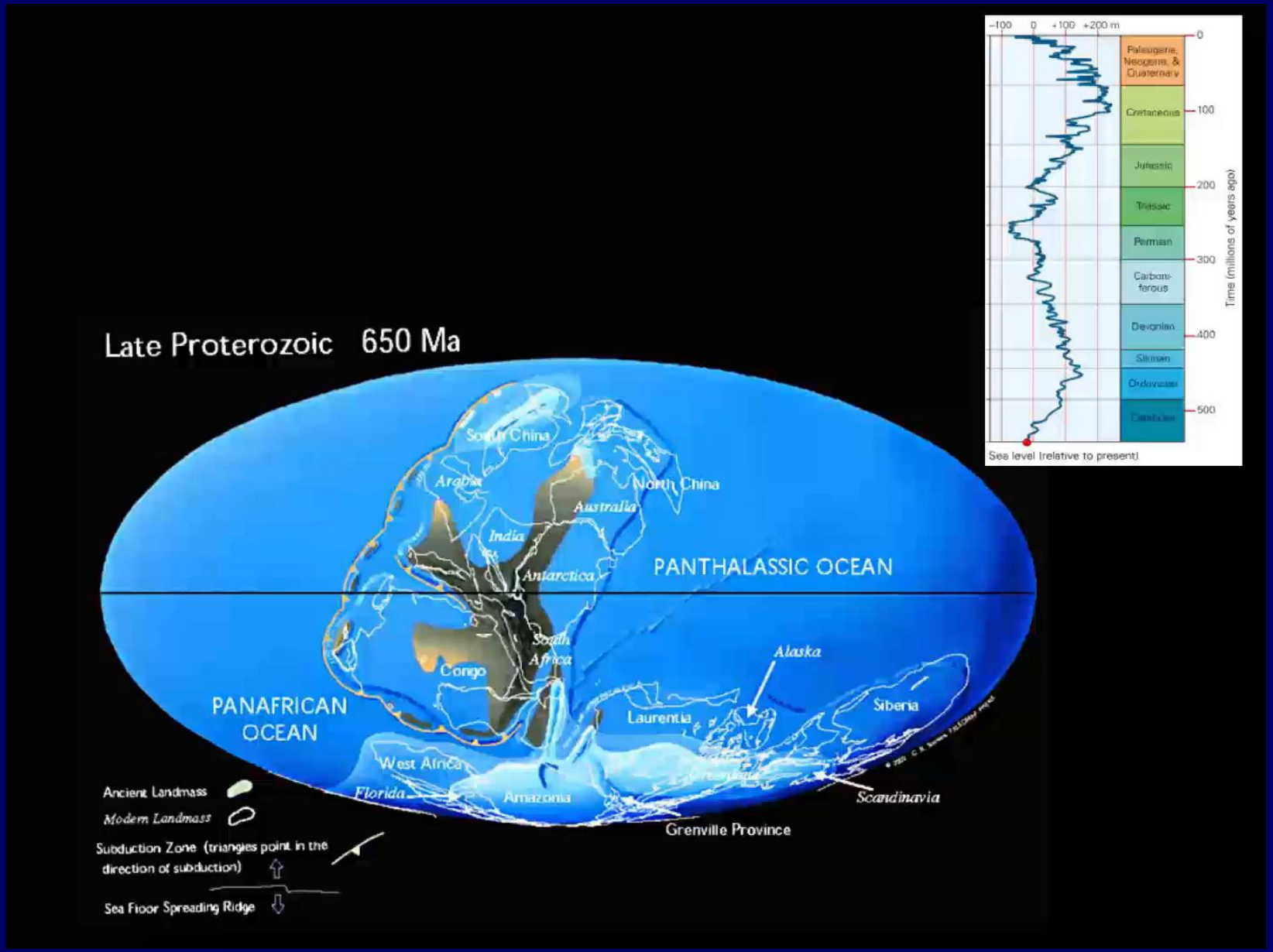
Requires:

- Barrier (to allow water to evaporate to high salinities)
- Mechanism to replenish brine
- Accommodation (space)





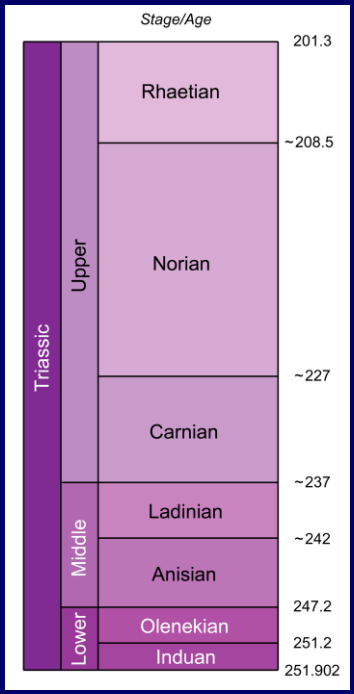
Phanerozoic tectonic plate motions and global seal level

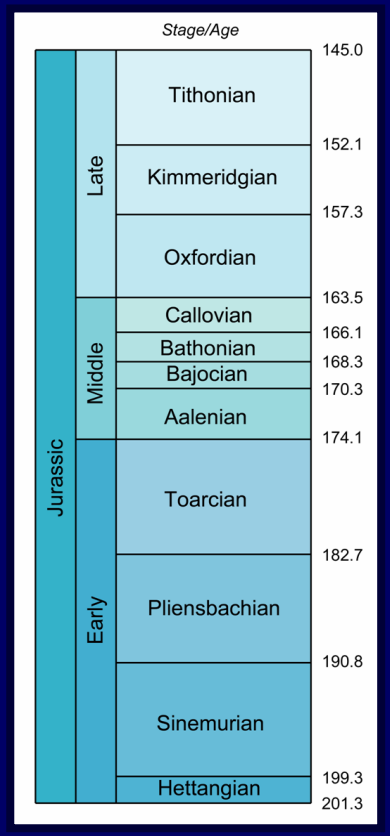


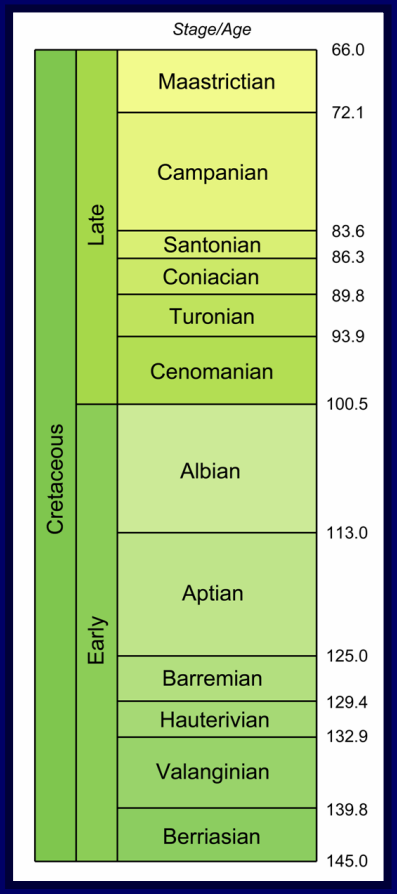
The Mesozoic Era: Legacy of a supercontinent

Mesozoic: middle life

- Permian-Triassic boundary: began with largest crash in global biodiversity of past 500 my, redirected course of evolution during the Mesozoic and Cenozoic eras
- Three periods:







Triassic = 50.6 myr duration
Jurassic = 56.3 myr duration
Cretaceous = 79 myr duration

The Triassic is named after three distinct rock layers (*tri* meaning 'three') found throughout northwestern Europe

The Jurassic is named after the Jura Mountains along the border between Switzerland and France

The Cretaceous is named from 'Creta' which is Latin for 'chalk'; Cretaceous = Chalk-bearing